

Eigenvalues of Tetrahedral sums of Hermitian matrices

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February 14, 2025

Abstract

TC: This is the MAIN draft for the TETRASPECTRA project. Older drafts are located in the folder titled “old” and will NOT compile properly.

Horn’s problem is concerned with characterizing the eigenvalues (α, β, γ) of a triple of Hermitian matrices (A, B, C) satisfying the constraint $A+B = C$. The development of a complete solution to Horn’s problem, due in large part to breakthroughs by A. Klyachko, A. Knutson, and T. Tao, has led to deep connections between eigenvalues of Hermitian matrices, tensor product invariants of matrix groups & Littlewood-Richardson coefficients, geometric invariant theory and the intersection theory of Schubert varieties via cohomology.

Here, we define and investigate a generalization of Horn’s problem which we refer to as the *tetrahedral Horn’s problem*. The tetrahedral Horn’s problem is concerned with characterizing the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying the constraints $A+B = C$, $B+D = F$, $D+C = E$ and $A+F = E$. In this paper, we describe the scaling, shifting and oriented tetrahedral symmetries of the tetrahedral Horn’s problem, as well as conjecture the existence of additional symmetries generalizing the Regge symmetries present in the case of 2×2 Hermitian matrices. Furthermore, unlike the solution to Horn’s problem, we show that the space of admissible sextuples of eigenvalues is neither convex nor a polytope. We demonstrate this qualitative difference explicitly in the case of 2×2 Hermitian matrices.

For the general case of $n \times n$ Hermitian matrices, we collect a handful of previously known matrix inequalities for sums of positive semidefinite matrices and reinterpret them as necessary conditions for the tetrahedral Horn’s problem. For a complete characterization, we provide a direct proof, using eigenvalue estimation techniques from quantum information theory, that the admissibility of a given sextuple of eigenvalues is encoded in the semiclassical asymptotics of the generalized Wigner $6j$ symbols for the unitary group $U(n)$. Our result generalizes the well-known correspondence between admissible eigenvalues for Horn’s problem and Littlewood-Richardson coefficients, and partially generalizes the well-known semiclassical analysis of J. Roberts for the case of $n = 2$.

Finally, we prove an upper bound on general $6j$ symbols using the Cauchy-Schwarz inequality which, in the $n \rightarrow \infty$ asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

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1 Introduction

There are a number of operations involving the transformation of Hermitian matrices under which it is possible to track the effect had of the corresponding sets of eigenvalues. For example, the eigenvalues, $\text{eig}(X)$, of a Hermitian matrix, X , are invariant under conjugation by unitary matrices, $\text{eig}(X) = \text{eig}(UXU^\dagger)$, covariant under taking positive powers $\text{eig}(X^p) = \text{eig}(X)^p$, covariant under multiplication by real numbers, $\text{eig}(rX) = r \text{eig}(X)$, and covariant under the addition of scalar matrices, $\text{eig}(X + rI) = \text{eig}(X) + (r, \dots, r)$. The operation of adding together Hermitian matrices is not an operation of this type. Specifically, the eigenvalues $\gamma = \text{eig}(A + B)$ of the sum of two matrices cannot be determined from the eigenvalues $\alpha = \text{eig}(A)$ and $\beta = \text{eig}(B)$ of the summands *alone*. Stated differently, to determine the eigenvalues involved, additional knowledge about the eigenvectors involved is typically required. Nevertheless, there are numerous mathematical situations wherein knowledge about the eigenvectors involved is lacking, yet it remains desirable to determine the set of admissible eigenvalues of a sum given the eigenvalues of the summands.

The problem of determining the set (α, β, γ) of admissible eigenvalues has been investigated for over a century yet has come to be known as *Horn's problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. These inequalities, now known as the *Horn inequalities* describe a polytope of admissible values for (α, β, γ) [Hor62]. In 1998, Klyachko provided a solution to Horn's problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98].

For applications of Horn's problem to quantum information theory and Nielsen's theorem [Nie99], see the article due to Daftuar and Hayden [DH05].

In parallel to Horn's problem, numerous variations and extensions of Horn's problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00]. This correspondence between the singular values of a product and the eigenvalues of sum has been generalized substantially by Alekseev, Meinrenken, and Woodward [AMW01]. For a pedagogical and historical introduction to Horn's problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00].

Recent developments surrounding Horn's problem not covered by the aforementioned reviews include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues of the sum given random matrices with fixed spectra in a series of papers by Coquereaux and Zuber [CZ18; CZ18; CMZ19; Zub18].

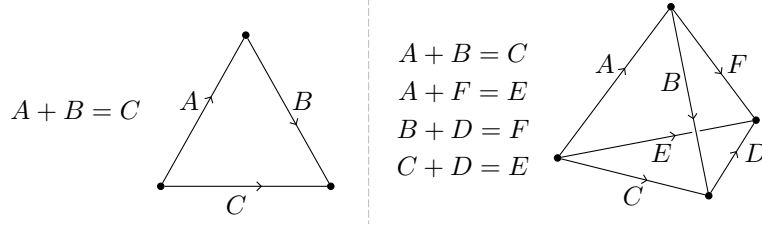


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. [fig:tetrahedron_labelling_scheme](#)

In this paper, we consider a generalization of Horn's problem previously considered only briefly in a paper due to Christandl, Şahinoğlu, and Walter [CŞW18]. While Horn's problem aims to characterize the eigenvalues (α, β, γ) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization consider here aims to characterize the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints: $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. These constraints arise naturally from building a kind of tetrahedral arrangement of Hermitian matrices inside the vector space of Hermitian matrices according to Fig. 1. Consequently, throughout the paper, we refer to this generalization of Horn's problem as the *tetrahedral Horn's problem* and, by contrast, Horn's original problem, involving the sum of just two Hermitian matrices, as the *triangular Horn's problem*.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless¹ Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. As we shall see in Example 7, this correspondence generalizes in the sense that the admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the $SU(2)$ Wigner $6j$ symbols [Rob99], it leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices. In Example 7, we will see that the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is provided by the positivity of so-called Cayley-Menger determinant [SJS18].

Although we are ultimately interested in tackling the tetrahedral Horn's problem for $n \times n$ Hermitian matrices with $n > 2$, the $n = 2$ case already demonstrates a strong qualitative difference between the triangular and tetrahedral variants of Horn's problem. In particular, while the triangular Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is in general *not a polytope*.

TODO: Finish the introduction

- Talk about connections between Horn's problem & Schubert calculus
- talk about semiclassical asymptotics of $6j$ symbols
- talk about entropy cones and open problems regarding entropy cones
- talk about known submodular inequalities for PSD matrices
- describe the scope & results of the paper (more detailed abstract)
- describe structure of the paper

Hyperbolic polynomials, Horn's problem, Lax conjecture [Ser09; Spe05].

¹One can always assume, without loss of generality, that the Hermitian matrices involved are traceless due to a simple shifting symmetry which is described in Section 2.3.

2 Tetrahedral eigenvalues

sec:tet_eig

2.1 The tetrahedral Horn problem

prob:tri_problem

Problem 1 (Trigonal Horn problem). A triple $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple (A, B, C) of $d \times d$ Hermitian matrices satisfying (i) the linear constraint

{eq:tri_linear_constraints}

$$A + B = C, \quad (1)$$

and (ii) the three eigenvalue constraints

{eq:tri_eigenvalue_constraints}

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma. \quad (2)$$

The problem of determining the set of trigonal eigenvalues (α, β, γ) will be referred to as the *trigonal Horn problem*.

prob:tet_problem

Problem 2 (Tetrahedral Horn problem). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^n)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

{eq:tet_linear_constraints}

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad (3)$$

and (ii) the six eigenvalue constraints

{eq:tet_eigenvalue_constraints}

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma, \quad \text{eig}(D) = \delta, \quad \text{eig}(E) = \epsilon, \quad \text{eig}(F) = \phi. \quad (4)$$

The problem of determining the set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the trigonal Horn problem in a variety of ways. First and foremost, the trigonal Horn problem is a special case of the tetrahedral Horn problem.

Proposition 3. A triplet (α, β, γ) of eigenvalues is trigonal if and only if the sextuple $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ is tetrahedral.

Proof. If (α, β, γ) are the eigenvalues of (A, B, C) satisfying $A+B = C$, then one can see that $(A, B, C, 0, C, B)$ satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, 0, \gamma, \beta)$. Conversely, if $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ are tetrahedral then it is straightforward to see that (α, β, γ) are trigonal. $\square$

More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of the trigonal Horn problem subject to additional matching constraints.

lem:four_faces

Lemma 4. A sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is tetrahedral only if each of

{eq:four_faces}

$$(\alpha, \beta, \gamma), \quad (\beta, \delta, \phi), \quad (\gamma, \delta, \epsilon), \quad (\alpha, \phi, \epsilon), \quad (5)$$

are trigonal.

Proof. The claim follows from a relaxation of the conditions defining the tetrahedral Horn problem. In particular, there exists a sextuple (A, B, C, D, E, F) of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying Eq. (3) only if there exists twelve Hermitian matrices

{eq:twelve_fold}

$$(A, A', B, B', C, C', D, D', E, E', F, F') \quad (6)$$

with eigenvalues $(\alpha, \alpha, \beta, \beta, \gamma, \gamma, \delta, \delta, \epsilon, \epsilon, \phi, \phi)$ satisfying

$$A + B = C, \quad B' + D = F, \quad C' + D' = E, \quad A' + F' = E'. \quad (7)$$

Now notice that each of the four linear constraints above depend on a disjoint subset of the twelve matrices in Eq. (6) and therefore the respective eigenvalues are independently constrained. In other words, this relaxation of the tetrahedral Horn problem is equivalent to the statement that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues only if each of four triplets in Eq. (5) are trigonal. $\square$

2.2 Low dimensional solutions

`exam:one_dim_soln`

Example 5 (Tetrahedral Horn's problem for $n = 1$). Perhaps the simplest example of the tetrahedral Horn's problem occurs when $n = 1$; i.e., when Hermitian matrices and their eigenvalues are identified with real numbers. In this case, the tetrahedral Horn's problem reduces to the problem of which sextuples of real numbers (a, b, c, d, e, f) simultaneously satisfy

$$a + b = c, \quad b + d = f, \quad d + c = e, \quad a + f = e, \quad (8)$$

`{eq:one_dim_tet_conds}`

or equivalently, are such that $(a, b, c, d, e, f) \in \ker \Gamma$ where Γ is the 4×6 matrix of coefficients encoding the linear system from Eq. (8):

$$\Gamma := \begin{pmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}. \quad (9)$$

`{eq:tet_coeff_matrix}`

It can be shown that the above coefficient matrix has a three dimensional kernel given by

$$(a, b, c, d, e, f) = (x, y, x + y, z, x + y + z, y + z). \quad (10)$$

`{eq:one_dim_soln}`

for $(x, y, z) \in \mathbb{R}^3$.

Remark 6 (Traces are tetrahedral). In the original, trigonal version of Horn's problem, the eigenvalues (α, β, γ) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$|\alpha| + |\beta| = |\gamma|, \quad (11)$$

`{eq:tri_trace_cond}`

where $|\lambda| := \sum_i \lambda_i$ is shorthand for the sum of the eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = |\lambda|$ for any Hermitian matrix X with eigenvalues λ . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ necessarily satisfy four *trace conditions*:

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\delta| + |\gamma| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (12)$$

`{eq:tet_trace_conds}`

Comparing the tetrahedral trace conditions (Eq. (12)) with the $n = 1$ case of the tetrahedral Horn's problem (Eq. (3)), we see that a sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can only be tetrahedral if their traces, $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$, are tetrahedral.

`exam:d_2`

Example 7 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. According to Corollary 13, one can restrict their focus to the case of *traceless* 2×2 Hermitian matrices without loss of generality. In this context, the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (13)$$

`{eq:traceless_two_dim_tetra}`

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (14)$$

Under this identification, the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (15)$$

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently,

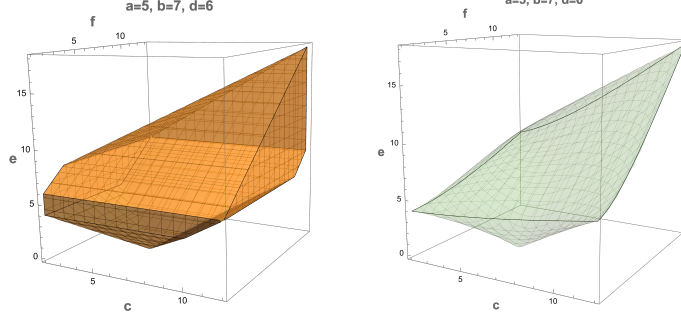


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (16). (Right) The set of all values (c, e, f) satisfying both Eq. (16) and Eq. (17).

in this setting, the tetrahedral Horn’s problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (16)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (17)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $D(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $D(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (16) and Eq. (17).

2.3 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Problem 2. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Problem 2 as stated. Later we will comment on the possibility of additional hidden symmetries not previously mentioned.

prop:scaling

Proposition 8 (Scaling symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, $(s\alpha, s\beta, s\gamma, s\delta, s\epsilon, s\phi)$, are tetrahedral if and only if the original, unscaled eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (3) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) ; for example, we have $sA + sB - sC = 0$ if and only if $s(A + B - C) = 0$ if and only if $A + B - C = 0$. $\square$

One may wonder with or not it is possible to extend the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues.

prop:inversion

Proposition 9 (Inversion symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let λ^* denote the inverted list of resorted and negated eigenvalues: $\lambda^* = (-\lambda_d \geq \dots \geq -\lambda_1)$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if $(\alpha^*, \beta^*, \gamma^*, \delta^*, \epsilon^*, \phi^*)$ are tetrahedral.*

Proof. The statement holds because (i) (A, B, C, D, E, F) satisfy Eq. (3) if and only if $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$ satisfy Eq. (3), and (ii) $\text{eig}(X) = \lambda$ if and only if $\text{eig}(-X) = \lambda^*$. $\square$

The inversion symmetry Proposition 9 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $\lambda \rightarrow \lambda^*$, or permutations of elements in the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$.

prop:tet-symmetry

Proposition 10 (Oriented Tetrahedral symmetry). *The sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

{eq:gens_of_tet_sym}

$$(\alpha \phi \epsilon \delta^* \gamma \beta) \quad (\gamma \epsilon^* \delta^* \phi \beta \alpha^*) \quad (18)$$

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc} (\alpha \beta \gamma \delta \epsilon \phi) & (\alpha^* \gamma \beta \delta \phi \epsilon) & (\alpha \gamma^* \beta^* \delta^* \phi^* \epsilon^*) & (\alpha^* \beta^* \gamma^* \delta^* \epsilon^* \phi^*) \\ (\delta \beta \phi \alpha \epsilon \gamma) & (\beta^* \alpha^* \gamma^* \epsilon \delta \phi) & (\beta \alpha \gamma \epsilon^* \delta^* \phi^*) & (\delta^* \beta^* \phi^* \alpha^* \epsilon^* \gamma^*) \\ (\delta^* \epsilon \gamma \alpha^* \beta \phi) & (\phi^* \beta \delta^* \gamma^* \epsilon^* \alpha^*) & (\phi \beta^* \delta \gamma \epsilon \alpha) & (\delta^* \epsilon^* \gamma^* \alpha \beta^* \phi^*) \\ (\alpha \epsilon^* \phi^* \delta \beta^* \gamma^*) & (\epsilon \delta^* \gamma \beta^* \alpha \phi^*) & (\epsilon^* \delta \gamma^* \beta \alpha^* \phi) & (\alpha^* \epsilon \phi \delta^* \beta \gamma) \\ & (\gamma \beta^* \alpha \phi \epsilon \delta) & (\gamma^* \beta \alpha^* \phi^* \epsilon^* \delta^*) & \\ & (\alpha \phi \epsilon \delta^* \gamma \beta) & (\alpha^* \phi^* \epsilon^* \delta \gamma^* \beta^*) & \\ & & & \\ (\beta \phi^* \delta^* \epsilon \gamma \alpha) & (\beta \delta \phi \epsilon^* \alpha^* \gamma^*) & (\beta^* \delta^* \phi^* \epsilon \alpha \gamma) & (\beta^* \phi \delta \epsilon^* \gamma^* \alpha^*) \\ (\gamma \alpha^* \beta \phi^* \delta^* \epsilon^*) & (\delta \phi^* \beta^* \alpha^* \gamma^* \epsilon^*) & (\delta^* \phi \beta \alpha \gamma \epsilon) & (\gamma^* \alpha \beta^* \phi \delta \epsilon) \\ (\epsilon \gamma^* \delta \beta \phi \alpha^*) & (\phi \epsilon^* \alpha^* \gamma \beta \delta^*) & (\phi^* \epsilon \alpha \gamma^* \beta^* \delta) & (\epsilon^* \gamma \delta^* \beta^* \phi^* \alpha) \\ (\phi \alpha \epsilon \gamma^* \delta \beta^*) & (\epsilon^* \alpha \phi^* \beta \delta^* \gamma) & (\epsilon \alpha^* \phi \beta^* \delta \gamma^*) & (\phi^* \alpha^* \epsilon^* \gamma \delta^* \beta) \\ (\phi^* \delta \beta^* \gamma \alpha \epsilon) & (\delta^* \gamma^* \epsilon^* \alpha \phi^* \beta^*) & (\delta \gamma \epsilon \alpha^* \phi \beta) & (\phi \delta^* \beta \gamma^* \alpha^* \epsilon^*) \\ (\epsilon^* \phi \alpha^* \beta^* \gamma^* \delta) & (\gamma^* \epsilon \delta \phi^* \beta^* \alpha) & (\gamma \epsilon^* \delta^* \phi \beta \alpha^*) & (\epsilon \phi^* \alpha \beta \gamma \delta^*) \\ (\gamma^* \delta^* \epsilon^* \phi \alpha^* \beta) & & & (\gamma \delta \epsilon \phi^* \alpha \beta^*) \\ (\beta^* \gamma \alpha \epsilon^* \phi^* \delta^*) & & & (\beta \gamma^* \alpha^* \epsilon \phi \delta) \end{array} \quad (19)$$

{eq:all_of_tet_sym}

Proof. To begin, notice that each of the operations on $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ appearing in Eq. (18) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (3). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues $(\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$) is also tetrahedral because

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned} \quad (20)$$

{eq:signed_perm_rows}

Moreover, as the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$ is invertible, one also has that if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is not tetrahedral, then neither is $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$. An analogous argument holds for the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 11. The group of order 48 appearing in Proposition 10 may be viewed as a subgroup of the group of all $6! \times 2^6 = 46,080$ possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Moreover, this subgroup is maximal in the sense that it is the maximal subgroup of permutations and negations of the columns of coefficient matrix Γ (appearing in Eq. (9)) which is equivalent to a permutation and negation of the rows of Γ (as in Eq. (20)). It can also be shown that this group of symmetries is isomorphic to the signed permutation group of three elements, i.e.,

$(\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes S_3$. In order to see this, it is helpful to arrange the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ into a 2×3 grid using the following identification:

$$(\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) \cong \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix}. \quad (21)$$

Then one can see that each $\mathbb{Z}_2$ factor in the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \alpha & \epsilon^* & \phi^* \\ \delta & \beta^* & \gamma^* \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta & \beta & \phi \\ \alpha & \epsilon & \gamma \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta^* & \epsilon & \gamma \\ \alpha^* & \beta & \phi \end{pmatrix}. \quad (22)$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \quad \begin{pmatrix} \beta & \gamma^* & \alpha^* \\ \epsilon & \phi & \delta \end{pmatrix} \quad \begin{pmatrix} \gamma^* & \alpha & \beta^* \\ \phi & \delta & \epsilon \end{pmatrix} \\ \begin{pmatrix} \alpha^* & \gamma & \beta \\ \delta & \phi & \epsilon \end{pmatrix} \quad \begin{pmatrix} \beta^* & \alpha^* & \gamma^* \\ \epsilon & \delta & \phi \end{pmatrix} \quad \begin{pmatrix} \gamma & \beta^* & \alpha \\ \phi & \epsilon & \delta \end{pmatrix} \quad (23)$$

Moreover, the above group of symmetries admits a finite presentation with respect to generators $g_1 \simeq (\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$ and $g_2 \simeq (\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ as a free group over g_1, g_2 subject to the relations

$$\{g_1^2, g_2^4, (g_1 g_2)^6, (g_2 g_1 g_2 g_1 g_2)^2\}. \quad (24)$$

prop:shifting

Proposition 12 (Shifting symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda + x := (\lambda_1 + x \geq \dots \geq \lambda_d + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{\alpha} &:= \alpha + x, & \tilde{\beta} &:= \beta + y, & \tilde{\gamma} &:= \gamma + x + y, & \tilde{\delta} &:= \delta + z, \\ \tilde{\epsilon} &:= \epsilon + x + y + z, & \tilde{\phi} &:= \phi + y + z. \end{aligned} \quad (25)$$

{eq:shifted_tet_eigenvalues}

The shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral if and only if the unshifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = \lambda$ and a constant $x \in \mathbb{R}$, the eigenvalues of $X + xI$ (where I is the identity matrix) are

$$\text{eig}(X + xI) = \lambda + x = (\lambda_1 + x \geq \dots \geq \lambda_d + x). \quad (26)$$

The statement then follows from Example 5 because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \quad (27)$$

satisfy the tetrahedral constraints in Eq. (3) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (3) $\square$

As a corollary of the shifting symmetries (Proposition 12) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 13) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 14).

cor:traceless

Corollary 13 (Traceless eigenvalues). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and for any list of n eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda^0 := \lambda - |\lambda|/n$ denote the traceless list of eigenvalues obtained by shifting λ by $-|\lambda|/n$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if (i) their traces $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$ are tetrahedral (they satisfy Eq. (12)) and (ii) $(\alpha^0, \beta^0, \gamma^0, \delta^0, \epsilon^0, \phi^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (12)) and applications of the shifting symmetry Proposition 12 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (28)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 12 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices (assuming some other basic inequalities hold). Note that in subsequent sections we will often explicitly assume the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace).

cor:psd

Corollary 14 (Positive eigenvalues). *If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$\gamma_d \geq \alpha_d + \beta_d, \quad \phi_d \geq \beta_d + \delta_d, \quad \epsilon_d \geq \delta_d + \gamma_d, \quad \epsilon_d \geq \alpha_d + \phi_d. \quad (29)$$

{eq:min_eig_ineqs}

Moreover, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a sextuple of candidate eigenvalues satisfying Eq. (29), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying

$$x > -\alpha_d, \quad y > -\beta_d, \quad z > -\delta_d, \quad (30)$$

{eq:positive_shift}

the resulting shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ (given by Eq. (25)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (29) follows from four applications of the well-known inequality $\gamma_d \geq \alpha_d + \beta_d$ for the trigonal Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue λ_n of a matrix X is characterized by a minimization: $\lambda_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$\gamma_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = \alpha_d + \beta_d. \quad (31)$$

Four applications of this inequality produces Eq. (29). Now if (x, y, z) satisfy Eq. (30), we have for all j

$$\alpha_j + x \geq \alpha_d + x > 0, \quad (32)$$

meaning $\alpha + x = (\alpha_1 + x \geq \dots \geq \alpha_d + x)$ is strictly positive. Similar statements hold for $\beta, \gamma, \delta, \epsilon$ and ϕ ; for instance, for all j ,

$$\epsilon_j + x + y + z \geq \epsilon_d + x + y + z > \epsilon_d - \alpha_d - \beta_d - \delta_d \geq 0, \quad (33)$$

meaning $\epsilon + x + y + z$ is strictly positive also. $\square$

Remark 15. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; MS18; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (34)$$

Combining this symmetry with the permutational symmetries given by Proposition 10 (and noting that for the eigenvalues λ of traceless 2×2 Hermitian matrices, $\lambda = (x, -x)$, we have $\lambda^* = \lambda$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (35)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (36)$$

It has also been shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dhen invariants [AI19]. Whether there exists analogous types of symmetries for the tetrahedral Horn's problem for $n \times n$ matrices when $n \geq 3$ remains an open problem [MS18].

2.4 Eigenvalues of blocks of positive semidefinite matrices

[sec:blocks_of_psd_matrices](#)

The purpose of this section is to reformulate both the trigonal and tetrahedral Horn's problem as a instance of a slightly more general kind of eigenvalue problem involving positive semidefinite matrices. Throughout this section, we will restrict our focus to the case of positive semidefinite matrices and their eigenvalues. As a reminder we note that this can be done without loss of generality due to Corollary 14.

Given a positive semidefinite matrix, X , with eigenvalues $\text{eig}(X) = \chi = (\chi_1 \geq \dots \geq \chi_n \geq 0)$ and eigen-decomposition $X = \sum_{i=1}^n \chi_i \Pi_i$, one can define $X^{\frac{1}{2}}$ by $X^{\frac{1}{2}} := \sum_{i=1}^n \chi_i^{\frac{1}{2}} \Pi_i$. Inspired by [Bha97, Theorem IV.2.13], we obtain the following fundamental lemma relating the eigenvalues γ of a sum $C = A + B$ of $n \times n$ Hermitian matrices to the eigenvalues of a larger $2n \times 2n$ matrix M built from A and B .

[lem:trigonal_block](#)

Lemma 16. *Let (A, B, C) be a triple of $n \times n$ Hermitian matrices with eigenvalues (α, β, γ) such that $C = A + B$. Let C' be the $2n \times 2n$ Hermitian matrix*

$$C' = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix}. \quad (37)$$

Then $C' \geq 0$ is positive semidefinite, has rank at most n and has eigenvalues given by γ

$$\text{eig}(C') \simeq \text{eig}(A + B) = \text{eig}(C) = \gamma. \quad (38)$$

Proof. That C' is positive semidefinite and has rank at most n follows because C' can be decomposed as $C' = T^* T$ where T is the $n \times 2n$ block matrix

$$T = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix}. \quad (39)$$

That C' and $A + B$ share the same eigenvalues (up to padding by n zeros) follows because for any rectangular matrix T , the eigenvalues of $T^* T$ and $T T^*$ are equal (up to padding by zeros) and in this instance

$$T^* T = \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix} = C' \quad T T^* = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} = A + B. \quad (40)$$

□

From the perspective of Lemma 16, one can view the trigonal Horn problem involving the eigenvalues (α, β, γ) of $n \times n$ Hermitian matrices (A, B, C) satisfying $A + B = C$ as a special case of a more general eigenvalue problem concerned with the relations between the eigenvalues of $2n \times 2n$ Hermitian matrices Σ of the form

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad (41)$$

[eq:block_ABX](#)

and the eigenvalues of its two $n \times n$ diagonal blocks Z and Z . It is known, by [HJ12, Theorem 7.7.9] or more generally [MKX19, Theorem 5.11] that if C' is a 2×2 block matrix such that $A', B' \geq 0$, then $C' \geq 0$ if and only if the off-diagonal block Z is of the form

$$Z = A'^{\frac{1}{2}} K B'^{\frac{1}{2}} \quad (42)$$

for some contraction matrix K (largest eigenvalue of $(K^* K)^{\frac{1}{2}}$ is at most unity). Additionally, when $Z = A'^{\frac{1}{2}} K B'^{\frac{1}{2}}$, one has the decomposition

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix} = \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} I & K \\ K^* & I \end{bmatrix} \begin{bmatrix} A'^{\frac{1}{2}} & 0 \\ 0 & B'^{\frac{1}{2}} \end{bmatrix}, \quad (43)$$

noting that K is a contraction if and only if the middle block matrix above is positive semidefinite [MKX19, Lemma 5.10].

Therefore, the special case covered by Lemma 16 corresponds to the case where the corresponding contraction matrix K is the $n \times n$ identity matrix $K = I$ (or more generally any contractive matrix satisfying $A'^{\frac{1}{2}} K B'^{\frac{1}{2}} = A'^{\frac{1}{2}} B'^{\frac{1}{2}}$). A simple observation which distinguishes between eigenvalue problem concerning

diagonal blocks and the eigenvalue problem concerning sums is simply that the rank of C' can in general be up to $2n$ whereas the rank of the sum $C = A + B$ is at most n . Despite this additional freedom, there remains non-trivial relations constraining the eigenvalues of C' and of the blocks A' and B' in terms of properties of Z (see [Gum+18; BL22]). Nevertheless, when the rank of C' is constrained to be at most n , we obtain the following theorem which provides an equivalent formulation of the trigonal Horn problem. *thm:trigonal_block*

Theorem 17. *Let (α, β, γ) be a triple of positive semidefinite eigenvalues of length at most n . The following are equivalent:*

- (i) *there exists a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$,*
- (ii) *there exists an $2n \times 2n$ Hermitian matrix,*

$$C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad (44)$$

such that $\text{eig}(A') = \alpha$, $\text{eig}(B') = \beta$, and $\text{eig}(C') = \gamma$.

Proof. Since we explicitly assume all of the eigenvalues in the triple (α, β, γ) are non-negative, the Hermitian matrices A, B, C , and A', B', C' involved in the statements above are necessarily positive semidefinite. The claim (i) $\implies$ (ii) follows from Lemma 16. That (ii) $\implies$ (i) follows by noticing that if C' has eigenvalues γ , then it has rank at most n . Therefore, as $C' \geq 0$, there exists a $n \times 2n$ block matrix T of the form

$$T = \begin{bmatrix} T_A & T_B \end{bmatrix} \quad (45)$$

such that $C' = T^*T$. The diagonal blocks of C' are therefore $A' = T_A^*T_A$, and $B' = T_B^*T_B$. As in the proof of Lemma 16, we note that T^*T and TT^* have the same eigenvalues up to padding by zeros. Therefore, by letting $A = T_A T_A^*$ and $B = T_B T_B^*$ we see that

$$\begin{aligned} \text{eig}(A) &= \text{eig}(T_A T_A^*) \simeq \text{eig}(T_A^* T_A) = \text{eig}(A') = \alpha, \\ \text{eig}(B) &= \text{eig}(T_B T_B^*) \simeq \text{eig}(T_B^* T_B) = \text{eig}(B') = \beta, \end{aligned} \quad (46)$$

while the eigenvalues of $C = A + B = T_A T_A^* + T_B T_B^*$ match the eigenvalues of C' because

$$\text{eig}(C) = \text{eig}(A + B) = \text{eig}(T_A T_A^* + T_B T_B^*) = \text{eig}(TT^*) \simeq \text{eig}(T^*T) = \text{eig}(C') = \gamma. \quad (47)$$

□

As a corollary of this correspondence between the trigonal Horn problem and the eigenvalues of blocks of 2×2 matrices, we obtain an equivalent formulation of the tetrahedral Horn problem. *cor:tetrahedral_block*

Corollary 18. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of positive semidefinite eigenvalues of length at most n . Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if there exists a $3n \times 3n$ matrix E' of the form*

$$E' = \begin{bmatrix} A' & Z & Y \\ Z^* & B' & X \\ Y^* & X^* & D' \end{bmatrix}, \quad \text{with blocks} \quad C' = \begin{bmatrix} A' & Z \\ Z^* & B' \end{bmatrix}, \quad F' = \begin{bmatrix} B' & X \\ X^* & D' \end{bmatrix} \quad \text{\textit{eq:block_3by3}} \quad (48)$$

such that

$$\text{eig}(A') = \alpha, \quad \text{eig}(B') = \beta, \quad \text{eig}(C') = \gamma, \quad \text{eig}(D') = \delta, \quad \text{eig}(E') = \epsilon, \quad \text{eig}(F') = \phi. \quad \text{\textit{eq:block_eigenvalue_tet}} \quad (49)$$

Proof. The proof is analogous to the proofs of Lemma 16 and Theorem 17. First we prove the “only if” portion of the claim which is a corollary of Lemma 16. Suppose (A, B, C, D, E, F) is a tetrahedral sextuple of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \gamma)$. Let E' be given by

$$E' = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix}^* \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} & A^{\frac{1}{2}} D^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B & B^{\frac{1}{2}} D^{\frac{1}{2}} \\ D^{\frac{1}{2}} A^{\frac{1}{2}} & D^{\frac{1}{2}} B^{\frac{1}{2}} & D \end{bmatrix} \quad (50)$$

Then the eigenvalues of the corresponding principal submatrices of E' satisfy Eq. (49) by Lemma 16. Now we prove the “if” portion of the claim which is a corollary of Theorem 17. If E' is a matrix of form given by Eq. (48) with eigenvalues ϵ with length at most n , then E' is positive semidefinite with rank at most n . Therefore, there exists an $n \times 3n$ block matrix

$$T = \begin{bmatrix} T_A & T_B & T_D \end{bmatrix} \quad (51)$$

such that $E' = T^*T$ and thus

$$\begin{aligned} A' &= T_A^* T_A, & B' &= T_B^* T_B, & D' &= T_D^* T_D, \\ X &= T_B^* T_D, & Y &= T_A^* T_D, & Z &= T_A^* T_B. \end{aligned} \quad (52)$$

Now let (A, B, C, D, E, F) be given by

$$\begin{aligned} A &= T_A T_A^*, & B &= T_B T_B^*, & C &= T_A T_A^* + T_B T_B^*, \\ D &= T_D T_D^*, & E &= T_A T_A^* + T_B T_B^* + T_D T_D^*, & F &= T_B T_B^* + T_D T_D^*. \end{aligned} \quad (53)$$

From here, one may directly verify that these $n \times n$ Hermitian matrices satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ (using a similar argument presented in the proof of Theorem 17); i.e., one concludes that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral. $\square$

3 Inequalities for tetrahedral eigenvalues

[sec:ineqs](#)

The purpose of this section is to describe various inequalities which are necessarily satisfied by tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. The task of finding a complete set of inequalities which is both necessary and sufficient for $n \geq 3$ remains an open problem.

3.1 Horn’s inequalities

Recall that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral only if each of (α, β, γ) , (β, γ, ϕ) , $(\gamma, \delta, \epsilon)$, and (α, ϕ, ϵ) are trigonal (Lemma 4). Consequently, every inequality constraining trigonal eigenvalues can be lifted to produce a quadruplet of inequalities constraining the set of tetrahedral eigenvalues. These inequalities can be understood as generalizations of the triangle inequalities, Eq. (16), which hold for case where $n = 2$. Therefore, before proceeding, we briefly summarize the inequalities satisfied by trigonal eigenvalues (α, β, γ) . Of particular interest are Horn’s inequalities [Hor62] which are known to be necessary and sufficient [Kly98; KT99]. For a more complete historical account, the reader is directed to the survey articles [Bha01; Ful98]. In addition to the aforementioned trace condition, $\sum_{i=1}^n \alpha_i + \sum_{j=1}^n \beta_j = \sum_{k=1}^n \gamma_k$, one always has the following pair of elementary inequalities,

$$\alpha_1 + \beta_1 \geq \gamma_1, \quad \text{and} \quad \alpha_n + \beta_n \leq \gamma_n, \quad (54)$$

which constrain the largest and smallest eigenvalues respectively (note that under inversion symmetry, these inequalities are equivalent). The first somewhat non-trivial inequality, derived by Weyl in 1912, is now known as *Weyl’s inequality* and states for all i, j such that $i + j \leq n + 1$ [Wey12]²

$$\alpha_i + \beta_j \geq \gamma_{i+j-1}. \quad (55)$$

In 1949, Fan derived a majorization inequality constraining the first r largest eigenvalues [Fan49], and shortly thereafter Lidskii [Lid53] & Wielandt [Wie55] found proofs of another family of inequalities. Finally, in 1962, Horn determined a complete collection of inequalities for (α, β, γ) are trigonal for the case where $n = 3$ [Hor62]. Moreover, also in [Hor62], Horn provided an inductively defined set of subsets $I, J, K \subset \{1, \dots, n\}$ of equal cardinality and conjectured that inequalities of the form

$$\sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j \geq \sum_{k \in K} \gamma_k, \quad (56)$$

²For 2×2 matrices one can see that these necessary inequalities are also sufficient to ensure that (α, β, γ) are trigonal; upon shifting to traceless eigenvalues using Corollary 13, $\alpha = (a, -a)$, $\beta = (b, -b)$ and $\gamma = (c, -c)$, and these inequalities become equivalent to the triangle inequalities $c \leq a + b$ and $c \geq \max\{b - a, a - b\}$.

were both necessary and sufficient to ensure that (α, β, γ) are trigonal. While proofs for some of these inequalities were found (see e.g., [TF71; TF70]), it took until the late 1990s when the combined works of Klyachko [Kly98] and Knutson and Tao [KT99] confirmed Horn's conjecture.

For the purposes of characterizing the set of tetrahedral eigenvalues further, we will henceforth be exclusively concerned with inequalities constraining the set of tetrahedral eigenvalues which are not already implied by Horn's inequalities.

3.2 A simple quadratic inequality

In Section 2.3 we saw that the shifting symmetry implies, in a certain sense, one may assume without loss of generality that the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are all non-negative or even strictly positive. While almost all of the following inequalities require this positivity assumption, we note that they can be extended to eigenvalues with negative entries in all cases.

To begin, we provide a simple example of a non-linear inequality constraining tetrahedral eigenvalues. prop. simple quadratic

Proposition 19. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be non-negative tetrahedral eigenvalues. Then*

$$2 \sum_{j=1}^n \alpha_j \delta_{n-j+1} \leq \sum_{j=1}^n (\beta_j^2 + \epsilon_j^2 - \gamma_j^2 - \phi_j^2) \leq 2 \sum_{j=1}^n \alpha_j \delta_j. \quad (57)$$

Proof. Let (A, B, C, D, E, F) be the tetrahedral sextuple of positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying $A + B = C$, $B + D = F$, and $A + F = C + D = E$. Then the following identity holds:

$$\begin{aligned} & \text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \\ &= \text{Tr}[(B)^2] + \text{Tr}[(A + B + D)^2] - \text{Tr}[(A + B)^2] - \text{Tr}[(B + D)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[AB] + 2 \text{Tr}[BD] + 2 \text{Tr}[AD]) \\ &\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2 \text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[BD]) \\ &= 2 \text{Tr}[AD]. \end{aligned} \quad (58)$$

While the value of $\text{Tr}(AD)$ cannot, in general, be determined from the eigenvalues $\alpha = \text{eig}(A)$ and $\delta = \text{eig}(D)$, one can apply the von Neumann trace inequalities [MOA10, H.1.g & H.1.h]:

$$\sum_j \alpha_j \delta_{n-j+1} \leq \text{Tr}[AD] \leq \sum_j \alpha_j \delta_j, \quad (59)$$

where $\delta_{n-j+1} = -\delta_j^*$. The claim then follows because $\text{Tr}[X^2] = \sum_j \lambda_j^2$ for any Hermitian operator X with eigenvalues $\text{eig}(X) = \lambda$. $\square$

3.3 Strongly subadditive/Submodular inequalities

DISCLAIMER FROM TC: The remainder of this section is written with respect to the A, B, C convention instead of the A, B, D convention. When discussing the tetrahedral Horn problem and its relation to the ordinary Horn problem or when matching the problem to the properties of the $6j$ symbols, the A, B, D convention is preferred. However when talking about submodular/strong subadditivity inequalities and comparing with the literature, the A, B, C convention is preferred. Before finalizing the paper, it will be desirable to establish a globally consistent convention. An equivalent formulation is to consider eigenvalues of four PSD matrices W, X, Y, Z with eigenvalues $(\omega, \chi, \nu, \zeta)$ such that $X + Y = W + Z$ and $W \geq X \geq Z$ and $W \geq Y \geq Z$.

Another strategy for understanding eigenvalues of positive semidefinite matrices and their sums is to consider the expansion of tensors powers of sums of matrices in relation to the tensor powers of their summands.³ For instance, given a positive integer k and positive semidefinite A and B , using the non-commutative binomial expansion of $(A + B)^{\otimes k}$, it is not too difficult to show [BS15, Proposition 1.1]:

$$(A + B)^{\otimes k} \geq A^{\otimes k} + B^{\otimes k}. \quad (60)$$

³We will exploit a variation of this strategy repeatedly in Section 4.

This observation can be generalized to obtain [TCL11, Lemma 2.2].

prop:operator_ineq

Proposition 20. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A + B + C)^{\otimes k} + B^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k}. \quad (61)$$

For further generalizations concerning operator inequalities involving the sum $A + C$ (which is outside the scope of the tetrahedral Horn problem), see [Lin14; BS15; NS23]. From Proposition 20 one can obtain an inequalities pertaining only to the eigenvalues of $B, A + B, B + C$, and $A + B + C$ by pairing the operator inequality with any $U(n)$ -equivariant positive semidefinite operator acting on $(\mathbb{C}^n)^{\otimes k}$.

cor:schur_ineq

Corollary 21. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let s_λ be the Schur polynomial indexed by the Young diagram $\lambda' = (\lambda'_1 \geq \lambda'_2 \geq \dots \geq \lambda'_n)$ with size $|\lambda'| := \lambda'_1 + \dots + \lambda'_n = k$. Then*

{eq:schur_ineq}

$$s_\lambda(\text{eig}(A + B + C)) + s_\lambda(\text{eig}(B)) \geq s_\lambda(\text{eig}(A + B)) + s_\lambda(\text{eig}(B + C)). \quad (62)$$

Proof. Given a Young diagram λ' , let $P_{\lambda'}$ be the Hermitian projection operator onto the isotypic subspace $V_{\lambda'}^n \otimes W_{\lambda'} \subseteq (\mathbb{C}^d)^{\otimes k}$ (see Section 4 for details). Given positive semidefinite matrix $X \geq 0$, one can show that

$$\text{Tr}(P_{\lambda'} X^{\otimes k}) = \text{Tr}(P_{\lambda'} \text{diag}(\text{eig}(X))^{\otimes k}) = \dim(W_{\lambda'}) s_\lambda(\text{eig}(X)) \geq 0. \quad (63)$$

Applying the positivity preserving map $X \mapsto \text{Tr}(P_{\lambda'} X^{\otimes k})$ to Proposition 20 yields

$$\text{Tr}[P_{\lambda'}(A + B + C)^{\otimes k}] + \text{Tr}[P_{\lambda'} B^{\otimes k}] \geq \text{Tr}[P_{\lambda'}(A + B)^{\otimes k}] + \text{Tr}[P_{\lambda'}(B + C)^{\otimes k}] \quad (64)$$

and since λ' has at most n rows, we conclude $\dim(W_{\lambda'}) \neq 0$ and thus Eq. (62) holds. $\square$

In [PTZ14] used the technique behind Proposition 20 to prove analogous inequalities concerning so-called generalized matrix functions d_χ^H associated to irreducible characters χ of subgroups H of the symmetric group S_k [MM65; HM21]. Unfortunately, these generalized matrix functions, such as the permanent of a matrix, are in general not invariant under unitary conjugation and thus depend on more just eigenvalues; the only exception is the determinant which is also a special case of Corollary 21 where $\lambda' = (1, \dots, 1)$, in which case $\text{Tr}(P_{(1, \dots, 1)} X^{\otimes n}) = \det(X)$ and thus one has

{eq:lin_det}

$$\det(A + B + C) + \det(B) \geq \det(A + B) + \det(B + C). \quad (65)$$

It is also possible to show a multiplicative variant of Eq. (65) also holds.

Proposition 22. *Let $A, B, C \geq 0$ be positive semidefinite matrices, then*

{eq:mul_det}

$$\det(A + B + C) \det(B) \leq \det(A + B) \det(B + C). \quad (66)$$

This inequality follows from [Bha97, Problem IX.8.11] whose proof is contained in the proof of an even stronger result proven later (Proposition 26). In the mean time, Eq. (66) can be viewed as a special case of a general result discovered by Sagnol [Sag13, Proposition 2.3].

thm:sagnol_ineq

Theorem 23. *Let $A, B, C \geq 0$ be positive semidefinite matrices. Let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ whose derivative f' is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

{eq:sagnol_ineq}

$$\text{Tr } f(A + B + C) + \text{Tr } f(B) \geq \text{Tr } f(A + B) + \text{Tr } f(B + C). \quad (67)$$

Proof. Here we sketch the proof in [Sag13, Section 2.1]. Let $F = \text{Tr} \circ f$. The claim then amounts to proving for all $A, B, C \geq 0$,

{eq:sagnol_diff}

$$\Delta_A \Delta_C F(B) = F(A + B + C) - F(A + B) - F(B + C) + F(B) \geq 0, \quad (68)$$

where Δ_X is the difference operator $[\Delta_X F](Y) := F(X + Y) - F(Y)$. Take $\varepsilon > 0$ be small and let $A = \varepsilon W$ for some non-zero positive semidefinite W . By standard continuity arguments (and the assumption that f is continuous), we may assume everything is positive definite in which case

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \Delta_{\varepsilon W} \Delta_C F(B) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (\Delta_C F(B + \varepsilon W) - \Delta_C F(B)) \quad (69)$$

$$= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} (F(\varepsilon W + B + C) - F(\varepsilon W + B) - F(B + C) + F(B)) \quad (70)$$

$$= \text{Tr}[(f'(B + C) - f'(B))W]. \quad (71)$$

Using the assumption that f' is an operator monotone (and the assumption of positive definiteness) together with $B + C \geq B$ (because $C \geq 0$), we have $f'(B + C) \geq f'(B)$ and thus $f'(B + C) - f'(B) \geq 0$ is a positive semidefinite operator. Therefore, the function $\Delta_C F(B)$ is non-decreasing under $B \mapsto A + B$ and thus Eq. (68) holds. $\square$

As immediate corollaries to Theorem 23, letting $f(x) = \log x$ and noting $\text{Tr}(\log X) = \log \det(X)$, one recovers Eq. (66). Moreover, taking $f(x) = -x^p$ for $p \in [0, 1]$, one obtains [BGS08, Lemma 5.3] which states for all $A, B, C \geq 0$ and $p \in [0, 1]$,

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (72)$$

On the other hand, letting $f(x) = x^p$ for $p \in (1, 2]$ yields for all $A, B, C \geq 0$ and $p \in (1, 2]$ the reverse

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (73)$$

Of particular interest to us in Section 4.4 is the function $f(x) = x \log x$ which implies for all $A, B, C \geq 0$ the inequality

$$S(A + B) + S(B + C) \geq S(B) + S(A + B + C) \quad (74)$$

where $S(X) = s(\text{eig}(X)) := -\sum_{i=1}^n \lambda_i(X) \log \lambda_i(X)$ is the *unnormalized* von Neumann entropy of X . Note that a similar inequality involving entropies of sums of positive semidefinite matrices (subject to additional eigenvalue constraints) has appeared in [CŠW18]. Moreover, while it can be shown that Eq. (74) is strictly tighter than the inequality in [CŠW18] for fixed matrices, it is also possible to obtain Eq. (74) from the result in [CŠW18] by downscaling the involved matrices in the sense of Proposition 19.

Remark 24. One consequence of Corollary 18, which relates the eigenvalues of sums of positive semidefinite matrices to the eigenvalues of diagonal blocks of larger positive semidefinite matrices with bounded rank, is that many of the aforementioned inequalities can now be seen as consequences of more general inequalities which constrain the eigenvalues of blocks.

For instance, the Kotljanskii determinant inequality [JB93] (in the notation of Corollary 18) implies for any 3×3 block matrix, that

$$\det(C') \det(F') \geq \det(E') \det(B'). \quad (75)$$

Using Corollary 18, the Kotljanskii determinant inequality becomes Eq. (66). More generally, Audenaert, Hiai, and Petz were interested in functions f which are strongly superadditive in the sense that for any 3×3 block matrices E' (again in the notation of Corollary 18) the following inequality holds [AHP10]:

$$\text{Tr } f(E') + \text{Tr } f(B') \geq \text{Tr } f(C') + \text{Tr } f(F'). \quad (76)$$

Their conclusion, later rediscovered by Friedland and Gaubert [FG13] was that it suffices for f' to be an operator monotone. In light of Corollary 18, the results of Audenaert, Hiai, and Petz and [FG13] then lead to an indirect proof of Theorem 23. In [LS14], Lewin and Sabin further proved that f' being an operator monotone is also *necessary* for Eq. (76) to hold for all 3×3 block matrices E' . Note that from the proof of Theorem 23 above, it is not too difficult to see that f' being an operator monotone is also necessary for Eq. (67) to hold (otherwise it becomes possible for Eq. (71) to become negative).

Remark 25. Much of the interest in characterizing the eigenvalues of principal submatrices of positive semidefinite matrices is closely related to the correspondence between positive semidefinite matrices and covariance matrices of Gaussian multivariate random variables [Lam+17]. From this perspective, principal submatrices become covariance matrices of Gaussian marginal distributions, and thus one may view the principal submatrix eigenvalue problem as a kind of “spectral” classical Gaussian marginal problem. Note that from this perspective, the differential entropies of the underlying Gaussian distributions are equal to logarithms of determinants up to an additive constant [Lam+17]. The Gaussian entropy region has been partially characterized by Shadbakht and Hassibi [SH11; HS08].

3.4 The Courtade-Sra determinant inequality

Many of the previously discussed inequalities constrain the eigenvalues of positive semidefinite matrices $B, A+B, B+C$ and $A+B+C$. Here we show how to improve the determinant inequality $\det(A+B)\det(B+C) \geq \det(A+B+C)\det(B)$ in a manner which also accounts for the determinants $\det(A)$ and $\det(C)$. This inequality exploits Minkowski’s determinant inequality for positive semidefinite $n \times n$ matrices A and B which reads (see [HJ85, Theorem 7.8.21] or [Bha97, Equation II.44]):

$$\det(A+B)^{\frac{1}{n}} \geq \det(A)^{\frac{1}{n}} + \det(B)^{\frac{1}{n}}. \quad (77)$$

The following inequality can be attributed to an information-theoretic result obtained by Courtade [Cou16]⁴, but the elementary proof we present here is actually due to Sra [Sra]. This inequality also appears in the recent review paper by Niculescu and Sra [NS23, Section 6.2].

prop:cs_ineq

Proposition 26 (Courtade-Sra determinant inequality). *Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices. Then*

$$\det(A+B)^{\frac{1}{n}} \det(B+C)^{\frac{1}{n}} \geq \det(A+B+C)^{\frac{1}{n}} \det(B)^{\frac{1}{n}} + \det(A)^{\frac{1}{n}} \det(C)^{\frac{1}{n}}. \quad (78)$$

eq:cs_ineq

Proof. By standard continuity arguments, it suffices to prove the inequality for $A, B, C > 0$ positive definite (applying the result to positive definite $(A + \varepsilon I, B + \varepsilon I, C + \varepsilon)$ and sending $\varepsilon \rightarrow 0$ yields the positive semidefinite claim). As B is positive definite, it is also invertible and we may define

$$X = B^{-\frac{1}{2}}AB^{-\frac{1}{2}}, \quad Y = B^{-\frac{1}{2}}CB^{-\frac{1}{2}}. \quad (79)$$

By dividing Eq. (78) by $\det(B)^{\frac{2}{n}} \neq 0$, it then suffices to prove

$$\det(I+X)^{\frac{1}{n}} \det(I+Y)^{\frac{1}{n}} \geq \det(I+X+Y)^{\frac{1}{n}} + \det(X)^{\frac{1}{n}} \det(Y)^{\frac{1}{n}}. \quad (80)$$

eq:cs_ineq_xy

Let $\lambda(X) = \text{eig}(X)$ and $\lambda(Y) = \text{eig}(Y)$. Then for any permutation $\pi \in S_n$ let

$$L_\pi = \text{diag}\{1 + \lambda_i(X) + \lambda_{\pi(i)}(Y)\}_{i=1}^n, \quad R_\pi = \text{diag}\{\lambda_i(X)\lambda_{\pi(i)}(Y)\}_{i=1}^n. \quad (81)$$

Then we have the identity

$$\det(I+X)\det(I+Y) = \prod_{i=1}^n (1 + \lambda_i(X)) \prod_{i=1}^n (1 + \lambda_i(Y)), \quad (82)$$

$$= \prod_{i=1}^n (1 + \lambda_i(X) + \lambda_{\pi(i)}(Y) + \lambda_i(X)\lambda_{\pi(i)}(Y)), \quad (83)$$

$$= \det(L_\pi + R_\pi) \quad (84)$$

Here we can apply Minkowski’s theorem to obtain

$$\det(L_\pi + R_\pi)^{\frac{1}{n}} \geq \det(L_\pi)^{\frac{1}{n}} + \det(R_\pi)^{\frac{1}{n}}. \quad (85)$$

⁴Note in [Cou17], Courtade conjectures a generalization of the Brunn-Minkowski inequality for bounded convex sets A, B, C in $\mathbb{R}^n$ of the form $V(A+B)^{\frac{1}{n}}V(B+C)^{\frac{1}{n}} \geq V(A+B+C)^{\frac{1}{n}}V(B)^{\frac{1}{n}} + V(A)^{\frac{1}{n}}V(C)^{\frac{1}{n}}$. See also [CC84].

By definition, we have $\det(R_\pi) = \det(X) \det(Y)$ for any permutation π , while a theorem do to Fiedler [Fie71] states (see also [Bha97, Exercise VI.7.2, Theorem VI.7.1])

$$\det(I + X + Y) \leq \max_{\pi} \prod_{i=1}^n (\lambda_i(I + X) + \lambda_{\pi(i)}(Y)) = \max_{\pi} \det(L_{\pi}). \quad (86)$$

Combining these bounds together yields Eq. (80) and thus the claim holds. $\square$

TC: On [mathoverflow](https://mathoverflow.net/questions/257562/a-determinant-inequality) <https://mathoverflow.net/questions/257562/a-determinant-inequality> Sra also proves

$$E_k(B)E_k(A + B + C) \leq E_k(A + B)E_k(B + C) \quad (87)$$

where $E_k(X) = e_k(\text{eig}(X))$ where E_k is the k th elementary symmetric polynomial.

The proof relies on a concavity result for hyperbolic polynomials due to Bauschke et al. [Bau+01]. An excellent crash course into Gårding's theory of hyperbolic polynomials (as well as some important results) is also [Bau+01]. For the particular case of the real vector space of $n \times n$ Hermitian matrices of dimension $n(n+1)/2$, the polynomial $X \mapsto \det(X)$ is hyperbolic in the direction of the identity matrix id_n (i.e., Hermitian matrices only have real eigenvalues). Moreover, the homogeneous polynomial $p_m(X) = \text{Tr}(P_{(1, \dots, 1)} X^{\otimes m})$ is the sum of the principal minors of X of size m and also equals the elementary symmetric polynomial of the eigenvalues of X , i.e.,

$$p_m(X) = E_m(\text{eig}(X)). \quad (88)$$

The homogeneous polynomial $p_m(X)$, like the determinant $\det(X) = p_n(X)$ itself, is also hyperbolic in the direction of the identity matrix id_n . In this context, together with the positivity $p_m(\text{id}_n) > 0$, Gårding's inequality generalizes the Minkowski determinant inequality and yields superlinearity of $p_m(X)^{\frac{1}{m}}$:

$$p_m(X + Y)^{\frac{1}{m}} \geq p_m(X)^{\frac{1}{m}} + p_m(Y)^{\frac{1}{m}} \quad (89)$$

for all X, Y positive semidefinite (the hyperbolicity cone for p_m). This inequality is due to Marcus and Lopes [ML57, Theorem 2], but it can also be proven using joint concavity of $(A_1, \dots, A_m) \mapsto A_1^{\frac{1}{m}} \otimes \dots \otimes A_m^{\frac{1}{m}}$ and thus concavity of $(A^{\frac{1}{m}})^{\otimes m}$ and thus by homogeneity the operator inequality $((A+B)^{\frac{1}{m}})^{\otimes m} \geq (A^{\frac{1}{m}})^{\otimes m} + (B^{\frac{1}{m}})^{\otimes m}$ ([Bha09, Exercise 4.2.9]). TC: I conjecture that Eq. (78) also holds for more general hyperbolic polynomials, although I do not have a proof yet.

4 Representation theory & eigenvalue estimation

`sec:rep_theory`

4.1 Schur-Weyl duality & eigenvalue estimation

Our main tool for connecting the eigenvalues of an $n \times n$ Hermitian matrix A to the representation theory of the unitary group $U(n)$ will be to examine the Schur-Weyl decomposition of the k -th tensor power $A^{\otimes k}$. To begin, we recall the Schur-Weyl decomposition of $(\mathbb{C}^n)^{\otimes k}$ as a representation of the group $S_k \times U(n)$.

Proposition 27 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times U(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda' \vdash_n k} W_{\lambda'} \otimes V_{\lambda'}^n \quad \text{\texttt{\{eq:schur_weyl_decomp\}}} \quad (90)$$

where the direct summands are indexed by Young diagrams λ' with k boxes and at most n rows, and where $W_{\lambda'}$ supports the irreducible representation of S_k and $V_{\lambda'}^n$ supports the irreducible representation of $U(n)$.

The dimension of the $V_{\lambda'}^n$ is given by Weyl's dimension formula [Hal08; IN66],

$$\dim(V_{\lambda'}^n) = \frac{\Delta_n(\tilde{\lambda}')}{\Delta_n(\rho)}, \quad \text{\texttt{\{eq:weyl_dim\}}} \quad (91)$$

whereas the dimension of $W_{\lambda'}$ may be given by [GW00, Corollary 9.1.5],

$$\dim(W_{\lambda'}) = \frac{|\lambda'|!}{\tilde{\lambda}'_1! \tilde{\lambda}'_2! \cdots \tilde{\lambda}'_n!} \Delta_n(\tilde{\lambda}') \quad \text{\textcolor{blue}{\{eq:frob_dim\}}} \quad (92)$$

where $\tilde{\lambda}' = \lambda' + \rho$, where $\rho = (n-1, n-2, \dots, 1, 0)$ is known as the Weyl vector and where $\Delta_n(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$ is the Vandermonde determinant. Note that the representation $W_{\lambda'}$ of S_k is independent of n ; indeed, the right hand side of Eq. (92) does not depend on n and coincides with the Hook-length formula [GW00, Corollary 9.1.6] (any value of n which is at least the depth of λ' suffices).

Definition 28. Given a non-zero positive semidefinite $n \times n$ matrix A , a positive integer x , and a Young diagram $\alpha' \vdash_n x$, let

$$\Gamma[A|\alpha'] := \text{Tr}[\underline{A}^{\otimes a} P_{\alpha'}], \quad (93)$$

where $\underline{A} = \frac{A}{\text{Tr}(A)}$ is the normalization of A and $P_{\alpha'}$ projects onto the isotypic component of $(\mathbb{C}^n)^{\otimes a}$ indexed by α' .

As $P_{\alpha'}$ is $U(n)$ -equivariant, one observes that $\Gamma[A|\alpha']$ depends only on the eigenvalues of A . The following result, known as spectrum or eigenvalue estimation, states that the map $\alpha' \mapsto \Gamma[A|\alpha']$ for A positive semidefinite is concentrated those Young diagrams α' whose shape matches the shape of the eigenvalues $\alpha = \text{eig}(A)$ [ARS88; KW01; HM02].

\textcolor{blue}{prop:eig_est}

Proposition 29 (Eigenvalue estimation). *Let A be a positive semidefinite matrix with eigenvalues $\alpha = \text{eig}(A)$ and let $\alpha' \vdash_n a$. Then*

$$\Gamma[A|\alpha'] \leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}'||\underline{\alpha})), \quad (94)$$

where $D(\underline{\alpha}'||\underline{\alpha})$ denotes the Kullback-Leibler divergence of $\underline{\alpha}'$ from $\underline{\alpha}$.

Proof. See [KW01] for a proof (noting that $\text{Tr}(\underline{A}) = 1$). □

\textcolor{blue}{prop:HS_measure}

Proposition 30. *Let $d\mu(A)$ denote the $U(n)$ -invariant Hilbert Schmidt probability measure over the set of $n \times n$ positive semidefinite trace one matrices and let x be a positive integer. Then*

$$\mathbb{E}_A[A^{\otimes a}] := \int d\mu(A) A^{\otimes k} = \sum_{\alpha' \vdash_n x} q_{\alpha'} P_{\alpha'} \quad \text{\textcolor{blue}{\{eq:HS_mean\}}} \quad (95)$$

where the coefficient $q_{\alpha'}$ equals

$$q_{\alpha'} = \frac{\dim(V_{\alpha'}^n)}{\dim(W_{\alpha'}) \dim(V_{(a)}^{n^2})} \quad \text{\textcolor{blue}{\{eq:HS_mean_coeff\}}} \quad (96)$$

Proof. See [ZS01] or [Wal14] for proof details. □

\textcolor{blue}{prop:HS_integral_uni}

Proposition 31. *Let A be a positive semidefinite $n \times n$ matrix such that $\text{Tr}(A) = 1$. Then*

$$\mathbb{E}_A \Gamma[A|\alpha'] := \int d\mu(A) \Gamma[A|\alpha'] = \frac{\dim(V_{\alpha'}^n)^2}{\dim(V_{(a)}^{n^2})}. \quad (97)$$

\textcolor{blue}{prop:completeness_rel_uni}

Proposition 32. *Let A be positive semidefinite $n \times n$ matrix with $\text{Tr}(A) = 1$ with eigenvalues $\alpha = \text{eig}(A)$ and rank $r(\alpha) = \text{rank}(A)$. Then for any positive integer a ,*

$$\sum_{\alpha' \vdash_{r(\alpha)} a} \Gamma[A|\alpha'] = 1, \quad (98)$$

where the sum is over Young diagrams α' with size a with depth at most $r(\alpha)$.

4.2 Littlewood-Richardson coefficients & trigonal eigenvalues

sec:lr_tri

The purpose of this section is to reveal how eigenvalue estimation can be used to prove a kind of correspondence between trigonal eigenvalue (α, β, γ) and Littlewood-Richardson coefficients $c_{\alpha', \beta'}^{\gamma'}$. Given a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$, by eigenvalue estimation we see that each of

$$\alpha' \mapsto \Gamma[A|\alpha'], \quad \beta' \mapsto \Gamma[B|\beta'], \quad \gamma' \mapsto \Gamma[C|\gamma'], \quad (99)$$

are respectively concentrated around Young diagrams $(\alpha', \beta', \gamma')$ whose normalized shapes $(\underline{\alpha'}, \underline{\beta'}, \underline{\gamma'})$ approximate the normalized eigenvalues $(\underline{\alpha}, \underline{\beta}, \underline{\gamma})$. As a result of this concentration, one can detect whether there exists Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ by examining if whether there exists an intersection of subspaces onto which $P_{\alpha'}$, $P_{\beta'}$ and $P_{\gamma'}$ project. To support this claim, we define the following quantity.

Definition 33. Given positive integers a and b , let $c = a + b$. Let $(\alpha', \beta', \gamma')$ be a triple of Young diagrams with sizes:

$$\alpha' \vdash a, \quad \beta' \vdash b, \quad \gamma' \vdash c, \quad (100)$$

{eq:three_yd_sizes}

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b}$ by

$$\Pi_{\alpha'} := P_{\alpha'} \otimes \text{id}^{\otimes b}, \quad \Pi_{\beta'} := \text{id}^{\otimes a} \otimes P_{\beta'}, \quad \Pi_{\gamma'} := P_{\gamma'}. \quad (101)$$

{eq:tri_extended_projs}

Given $n \times n$ positive semidefinite matrices A, B and $C = A + B$, let

$$\Gamma[A \ B \ C | \alpha' \ \beta' \ \gamma'] := \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b})(P_{\alpha'} \otimes P_{\beta'})P_{\gamma'}] \quad (102)$$

defn:lr_coeffs

Definition 34 (Littlewood-Richardson coefficients). Let $(\alpha', \beta', \gamma')$ be a triple of Young diagrams with sizes (a, b, c) where $c = a + b$ and let $V_{\alpha'}^n, V_{\beta'}^n, V_{\gamma'}^n$ be a triple of irreducible representations of $U(n)$. Let $C_{\alpha', \beta'}^{\gamma'}$ denote the multiplicity space of $U(n)$ -intertwiners from $V_{\gamma'}$ to $V_{\alpha'} \otimes V_{\beta'}$:

$$C_{\alpha', \beta'}^{\gamma'} := \text{Hom}_{U(n)}(V_{\gamma'}, V_{\alpha'} \otimes V_{\beta'}) \simeq \text{Inv}_{U(n)}(V_{\gamma'^*} \otimes V_{\alpha'} \otimes V_{\beta'}). \quad (103)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha', \beta'}^{\gamma'} := \dim(C_{\alpha', \beta'}^{\gamma'}). \quad (104)$$

The *Clebsch-Gordon isometry* is the $U(n)$ -equivariant isometric linear map

$$Y_{\alpha, \beta}^{\gamma} : C_{\alpha, \beta}^{\gamma} \otimes V_{\gamma} \rightarrow V_{\alpha} \otimes V_{\beta}, \quad (105)$$

sending $f \otimes v \in C_{\alpha', \beta'}^{\gamma'} \otimes V_{\gamma'}$ to $f(v) \in V_{\alpha'} \otimes V_{\beta'}$ ⁵. The Clebsch-Gordon isometries (and coisometries) will be depicted diagrammatically with multiplicity spaces given a distinguished style as in [AB68]:

$$\begin{array}{ccc} \begin{array}{c} V_{\alpha} \swarrow \searrow V_{\beta} \\ \circ \\ \nearrow C_{\alpha, \beta}^{\gamma} \searrow V_{\gamma} \end{array} & \equiv & \begin{array}{c} V_{\alpha} \swarrow \searrow V_{\beta} \\ \boxed{Y_{\alpha, \beta}^{\gamma}} \\ \nearrow C_{\alpha, \beta}^{\gamma} \searrow V_{\gamma} \end{array} \end{array} \quad \begin{array}{ccc} \begin{array}{c} C_{\alpha, \beta}^{\gamma} \swarrow \searrow V_{\gamma} \\ \circ \\ \nearrow V_{\alpha} \searrow V_{\beta} \end{array} & \equiv & \begin{array}{c} C_{\alpha, \beta}^{\gamma} \swarrow \searrow V_{\gamma} \\ \boxed{Y_{\alpha, \beta}^{\gamma, *}} \\ \nearrow V_{\alpha} \searrow V_{\beta} \end{array} \end{array} \quad (106)$$

TODO: Summarize [Chr08]. TODO: Briefly mention the direct and pedagogical Kempf-Ness style proofs of stronger results in [BVW18, Proposition 2.3] also using the strong duality result of [FW20, Theorem 1 & Proposition 3.1] and the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17].

⁵If $C_{\alpha', \beta'}^{\gamma'} = 0$, then $Y_{\alpha', \beta'}^{\gamma'}$ is identified with the zero map.

4.3 Generalized $6j$ symbols & tetrahedral eigenvalues

`defactsixgamma`

Definition 35. Given positive integers a, b , and d , let $c = a + b$, $f = b + d$, and $e = a + b + d$. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams with sizes:

$$\alpha' \vdash a, \quad \beta' \vdash b, \quad \gamma' \vdash c, \quad \delta' \vdash d, \quad \epsilon' \vdash e, \quad \phi' \vdash f. \quad (108)$$

`{eq:six_yd_sizes}`

Define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes a+b+d}$ by

$$\begin{aligned} \Pi_{\alpha'} &:= P_{\alpha'} \otimes \text{id}^{\otimes b} \otimes \text{id}^{\otimes d}, & \Pi_{\beta'} &:= \text{id}^{\otimes a} \otimes P_{\beta'} \otimes \text{id}^{\otimes d}, & \Pi_{\gamma'} &:= P_{\gamma'} \otimes \text{id}^{\otimes d}, \\ \Pi_{\delta'} &:= \text{id}^{\otimes a} \otimes \text{id}^{\otimes b} \otimes P_{\delta'}, & \Pi_{\epsilon'} &:= P_{\epsilon'}, & \Pi_{\phi'} &:= \text{id}^{\otimes a} \otimes P_{\phi'}. \end{aligned} \quad (109)$$

`{eq:tet_extended_projs}`

Given $n \times n$ positive semidefinite matrices A, B, D with $C = A + B$, $F = B + D$, and $E = A + B + D$, let

$$\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{bmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{bmatrix} = \text{Tr}[(\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}]. \quad (110)$$

Remark 36. In the analysis that follows, it will be convenient to avoid certain edge cases pertaining to the candidate sextuple of positive semidefinite eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Specifically, let the extremal traces be denoted by

$$\tau_{\min} := \min\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad \tau_{\max} := \max\{|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|\}, \quad (111)$$

and the extremal ranks be denoted by

$$r_{\min} := \min\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}, \quad r_{\max} := \max\{r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi)\}. \quad (112)$$

In order to avoid the edge case where one of the eigenvalues $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$ is totally zero, we sometimes demand that the minimum trace is strictly positive or equivalently that the minimum rank is at least one:

$$\tau_{\min} > 0, \quad r_{\min} \geq 1. \quad (113)$$

Another technical difficulty which complicates subsequent analysis is that the set of projection operators $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ defined by Eq. (109) is *almost* pairwise mutually commuting.

`lem:tet_com_rels`

Lemma 37. Every pair of projection operators in $\{\Pi_{\alpha'}, \Pi_{\beta'}, \Pi_{\gamma'}, \Pi_{\delta'}, \Pi_{\epsilon'}, \Pi_{\phi'}\}$ is commuting with only possible exception being $\Pi_{\gamma'}$ and $\Pi_{\phi'}$. Therefore, the operators $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ defined by

$$\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'} = \Pi_{\gamma'} \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\delta'} \Pi_{\epsilon'}, \quad \text{and} \quad \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} = \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Pi_{\delta'} \Pi_{\epsilon'}, \quad (114)$$

`{eq:commuting_extended_projs}`

are projection operators.

Proof. That all projection operators except the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ commute follows from elementary applications of Schur's lemma. The obstruction for the pair $\{\Pi_{\gamma'}, \Pi_{\phi'}\}$ is simply that $\Pi_{\gamma'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes b+d}$ and likewise $\Pi_{\phi'}$ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes a+b}$. $\square$

defn:sixj

Definition 38 (Generalized 6j operators). Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams with sizes (a, b, c, d, e, f) given by Eq. (108). Then we define the $U(n)$ -6j operator to be the operator⁶

$$\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} : C_{\alpha', \beta'}^{\gamma'} \otimes C_{\delta', \gamma'}^{\epsilon'} \rightarrow C_{\alpha', \phi'}^{\epsilon'} \otimes C_{\beta', \delta'}^{\phi'}, \quad (115)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} := \frac{1}{\dim(V_{\epsilon'})} \text{Tr}_{V_{\epsilon'}} [Y_{\alpha', \phi'}^{\epsilon' \dagger} (\text{id}_{V_{\alpha'}} \otimes Y_{\beta', \delta'}^{\phi' \dagger}) (Y_{\alpha', \beta'}^{\gamma'} \otimes \text{id}_{V_{\delta'}}) Y_{\gamma', \delta'}^{\epsilon'}], \quad (116)$$

eq:sixj

and depicted diagrammatically by

The following proposition is the analogue of Proposition 31.

prop:HS_integral_tet

Proposition 39. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes given by Eq. (108). Then

$$\begin{aligned} \mathbb{E}_{A,B,D} \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] &:= \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] \\ &= R_{\alpha' \beta' \delta'; \epsilon'} \left\| \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right\|_{\text{HS}}^2 \end{aligned} \quad (118)$$

eq:tet_mean_value

where $R_{\alpha' \beta' \delta'; \epsilon'}$ is a ratio of dimensions:

$$R_{\alpha' \beta' \delta'; \epsilon'} = \frac{\dim(V_{\alpha'}^n) \dim(V_{\beta'}^n) \dim(V_{\delta'}^n) \dim(V_{\epsilon'}^n)}{\dim(V_{(a)}^{n^2}) \dim(V_{(b)}^{n^2}) \dim(V_{(d)}^{n^2})}. \quad (119)$$

Proof. To prove Eq. (118), we use Proposition 30 and Lemma 37 to see that

$$\mathbb{E}_{A,B,D} \Gamma \left[\begin{array}{ccc|ccc} A & B & C & \alpha' & \beta' & \gamma' \\ D & E & F & \delta' & \epsilon' & \phi' \end{array} \right] = q_{\alpha'} q_{\beta'} q_{\delta'} \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}). \quad (120)$$

eq:tet_mean_step_1

We claim that the product $\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}$ is related to the generalized 6j operator $\left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\}$ in that there exists appropriately-typed isometries ι_I and ι_{II} such that

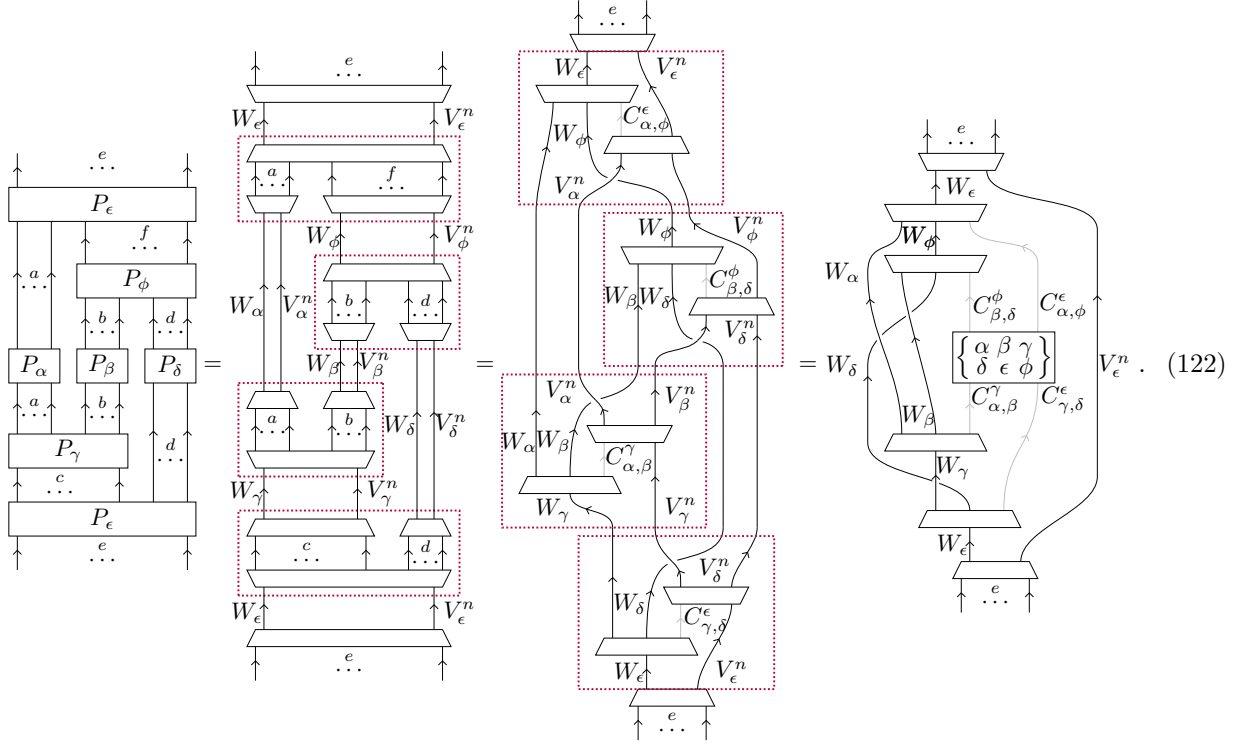
$$\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} = \iota_{II} \left(\text{id}_{V_{\epsilon'}} \otimes \text{id}_{W_{\alpha}} \otimes \text{id}_{W_{\beta}} \otimes \text{id}_{W_{\delta}} \otimes \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right) \iota_I^{\dagger} \quad (121)$$

eq:ud_sixj_prod_proj_ise

The form of these isometries can be deduced by applying Eq. (107) four times in the following diagrammatic

⁶It is important to note that there does not seem to exist a universal convention concerning how 6j symbols are normalized; Coquereaux discusses this problem in [Coq23].

calculation:



From Eq. (121) we conclude

$$\text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'}) = \dim(V_{\epsilon}^n) \dim(W_{\alpha}) \dim(W_{\beta}) \dim(W_{\delta}) \left\| \left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta' & \epsilon' & \phi' \end{matrix} \right\} \right\|_{\text{HS}}^{\text{eq:nd_six}_2 \text{prod_proj_hs_norm}}, \quad (123)$$

Combining Eq. (123) with Eq. (120) and Eq. (96) then yields Eq. (118). $\square$

The following proposition is the analogue of Proposition 32.

prop:completeness_rel_tet

Proposition 40. *Let (A, B, C, D, E, F) be tetrahedral positive semidefinite matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Then*

$$\sum_{\substack{\alpha' \beta' \gamma' \\ \delta' \epsilon' \phi'}} \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} = 1. \quad (124)$$

eq:tet_completeness

where the sum is taken over Young diagrams with the following sizes:

$$\alpha' \vdash_{r(\alpha)} a, \quad \beta' \vdash_{r(\beta)} b, \quad \gamma' \vdash_{r(\gamma)} c, \quad \delta' \vdash_{r(\delta)} d, \quad \epsilon' \vdash_{r(\epsilon)} e, \quad \phi' \vdash_{r(\phi)} f. \quad (125)$$

eq:six_yd_sizes_rank

Proof. To begin, note that Eq. (124) holds when the sum is taken over Young diagrams with unrestricted rank according to Eq. (108) because $\sum_{\lambda' \vdash_n \ell} P_{\lambda'} = \text{id}_n^{\otimes \ell}$ for all positive integers ℓ and $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$. That one can restrict to Young diagrams with depths bounded according to Eq. (108) follows because $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix} = 0$ whenever the depth of *any* of $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ exceeds the corresponding ranks $(r(\alpha), r(\beta), r(\gamma), r(\delta), r(\epsilon), r(\phi))$.⁷ $\square$

Our next goal will be to derive two upper bounds on the absolute value of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{matrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{matrix}$. Our first upper bound is *independent* of the given matrices and follows as a corollary of Lemma 37.

⁷This claim actually follows from Theorem 42 and will be proven shortly. Importantly, Theorem 42 does not rely on Proposition 40.

Corollary 41. Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of sizes given by Eq. (108). Then for all tetrahedral, non-zero positive semidefinite (A, B, C, D, E, F) , we have

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\infty}. \quad \text{feq:tet_gamma_size_upper_bound} \quad (126)$$

Proof. From Eq. (114) we have

$$\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} = \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (127)$$

Moreover, as $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$ is positive semidefinite trace one, and $\Pi_1 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ and $\Pi_2 := \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ are projection operators by Lemma 37, we have

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| = |\text{Tr}(\Pi_1 \Omega \Pi_2)| \leq |\langle v, \Pi_1 \Pi_2 v \rangle| \leq \|\Pi_1 \Pi_2\|_{\infty} = \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\infty}. \quad (128)$$

where the last equality follows from the isometry decomposition provided by Eq. (121). $\square$

Our second upper bound on the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ depends on the *eigenvalues* $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of the given tetrahedral matrices (A, B, C, D, E, F) where $C = A + B$, $F = B + D$ and $E = A + B + D$. This result proves that in the limit of large values for (x, y, z) , the magnitude of $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ is sharply concentrated around those sextuples of Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ which resemble the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of (A, B, C, D, E, F) .

thm:abd_dep_upper_bound

Theorem 42. Let everything be as in Definition 35 and let the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes given by

$$\begin{aligned} \alpha' \vdash a &= [k | \alpha|], & \beta' \vdash b &= [k | \beta|], & \gamma' \vdash c &= [k | \alpha|] + [k | \beta|], \\ \delta' \vdash d &= [k | \delta|], & \epsilon' \vdash e &= [k | \alpha|] + [k | \beta|] + [k | \delta|], & \phi' \vdash f &= [k | \beta|] + [k | \delta|], \end{aligned} \quad \text{feq:ceil_yd_sizes} \quad (129)$$

for some positive integer $k > \tau_{\min}^{-1}$ where $\tau_{\min}$ is the minimum trace. Then

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \leq p(k) \exp(-k I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad \text{feq:tet_gamma_upper_bound} \quad (130)$$

where $p(k)$ is a polynomial in k and where $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$ is a quantity which measures the deviation of the primed quantities from the unprimed quantities:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = |\alpha| D(\underline{\alpha}' || \underline{\alpha}) + |\beta| D(\underline{\beta}' || \underline{\beta}) + |\gamma| D(\underline{\gamma}' || \underline{\gamma}) + |\delta| D(\underline{\delta}' || \underline{\delta}) + |\epsilon| D(\underline{\epsilon}' || \underline{\epsilon}) + |\phi| D(\underline{\phi}' || \underline{\phi}). \quad (131)$$

Proof. The first portion of the proof does not depend on the value of k . For now, we suppose the Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ have sizes determined by Eq. (108) for x, y, z arbitrary positive integers. As in the proof of Corollary 41, we let $\Omega := \underline{A}^{\otimes a} \otimes \underline{B}^{\otimes b} \otimes \underline{D}^{\otimes d}$. Using the commutation relations provided by Lemma 37 we apply the Cauchy-Schwarz inequality to $\langle X, Y \rangle = \text{Tr}(X^* Y) \leq \text{Tr}(X^* X)^{\frac{1}{2}} \text{Tr}(Y^* Y)^{\frac{1}{2}}$ for $X = \Omega^{\frac{1}{2}} \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $Y = \Omega^{\frac{1}{2}} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ to obtain

$$\begin{aligned} \left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega), \end{aligned} \quad (132)$$

where the last inequality follows because $\Pi_{\lambda'} \leq \text{id}_n^{|\lambda'|}$ for each λ' in $\{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$.

Now, for each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the eigenvalue estimation technique (Proposition 29). For a detailed example, we will consider value of $\text{Tr}(\Pi_{\gamma'} \Omega)$:

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes d})(A^{\otimes a} \otimes B^{\otimes b} \otimes D^{\otimes d})]}{|\alpha|^a |\beta|^b |\delta|^d} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes a} \otimes B^{\otimes b})]}{|\alpha|^a |\beta|^b}. \quad (133)$$

Under the assumption that $C = A + B$, we have $C^{\otimes c} = (A + B)^{\otimes a+b}$ which implies

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) = \sum_{k=0}^c \binom{c}{k} \text{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes c-k})). \quad (134)$$

Here we have used the S_c -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq j \leq c$ that

$$\text{Tr}(P_{\gamma'} C^{\otimes c}) \geq \binom{c}{j} \text{Tr}(P_{\gamma'} (A^{\otimes j} \otimes B^{\otimes c-j})). \quad (135)$$

Letting $j = a$ and rearranging yields

$$\text{Tr}(P_{\gamma'} (A^{\otimes a} \otimes B^{\otimes b})) \leq \frac{a!b!}{c!} \text{Tr}(P_{\gamma'} C^{\otimes c}). \quad (136)$$

Combining this result with the spectrum estimation bound (Proposition 29) yields

$$\text{Tr}(\Pi_{\gamma'} \Omega) \leq \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \dim(V_{\gamma'}^n) \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})). \quad (137)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\begin{aligned} \text{Tr}(\Pi_{\alpha'} \Omega) &\leq \dim(V_{\alpha'}^n) \exp(-aD(\underline{\alpha}' \parallel \underline{\alpha})), \\ \text{Tr}(\Pi_{\beta'} \Omega) &\leq \dim(V_{\beta'}^n) \exp(-bD(\underline{\beta}' \parallel \underline{\beta})), \\ \text{Tr}(\Pi_{\gamma'} \Omega) &\leq \dim(V_{\gamma'}^n) \frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \exp(-cD(\underline{\gamma}' \parallel \underline{\gamma})), \\ \text{Tr}(\Pi_{\delta'} \Omega) &\leq \dim(V_{\delta'}^n) \exp(-dD(\underline{\delta}' \parallel \underline{\delta})), \\ \text{Tr}(\Pi_{\epsilon'} \Omega) &\leq \dim(V_{\epsilon'}^n) \frac{|\epsilon|^e}{|\alpha|^a |\beta|^b |\delta|^d} \frac{a!b!d!}{e!} \exp(-eD(\underline{\epsilon}' \parallel \underline{\epsilon})), \\ \text{Tr}(\Pi_{\phi'} \Omega) &\leq \dim(V_{\phi'}^n) \frac{|\phi|^f}{|\beta|^b |\delta|^d} \frac{b!d!}{f!} \exp(-fD(\underline{\phi}' \parallel \underline{\phi})). \end{aligned} \quad (138)$$

Our objective now will be to use the k -dependent values for (a, b, c, d, e, f) to upper bound the multinomial prefactors appearing in the bounds for γ' , ϵ' , and ϕ' above in terms of k . From Eq. (129), we obtain the following estimates:

$$\begin{aligned} k|\alpha| &\leq a < k|\alpha| + 1, & k|\beta| &\leq b < k|\beta| + 1, & k|\gamma| &\leq c < k|\gamma| + 2, \\ k|\delta| &\leq d < k|\delta| + 1, & k|\epsilon| &\leq e < k|\epsilon| + 3, & k|\phi| &\leq f < k|\phi| + 2. \end{aligned} \quad (139)$$

An application of Lemma 50 for binomial distributions ($m = 2$) implies the following bound

$$\frac{|\gamma|^c}{|\alpha|^a |\beta|^b} \frac{a!b!}{c!} \leq 12^2 (k|\gamma| + 2)^{\frac{1}{2}}. \quad (140)$$

Moreover, from Weyl's dimension formula, Eq. (91), we obtain [CM06, Equation 7]:

$$\dim(V_{\gamma'}^n) \leq (|\gamma'| + 1)^{n(n-1)/2} \leq (k|\gamma| + 3)^{n(n-1)/2}, \quad (141)$$

because $(\gamma'_i - i) - (\gamma'_j - j) \leq |\gamma'| + 1$ as well as $j - i \geq 1$. Altogether, we can relax the bound on $\text{Tr}(\Pi_{\gamma'} \Omega)$ to obtain

$$\text{Tr}(\Pi_{\gamma'} \Omega) \leq 12^2 (k|\gamma| + 3)^{\frac{n(n-1)}{2} + \frac{1}{2}} \exp(-k|\gamma| D(\underline{\gamma}' \parallel \underline{\gamma})). \quad (142)$$

A similar calculations for $\text{Tr}(\Pi_{\epsilon'} \Omega)$ yields (noting to apply Lemma 50 for trinomial distributions when $m = 3$) yields:

$$\text{Tr}(\Pi_{\epsilon'} \Omega) \leq 12^3 (k|\epsilon| + 4)^{\frac{n(n-1)}{2} + 1} \exp(-k|\epsilon| D(\underline{\epsilon}' \parallel \underline{\epsilon})). \quad (143)$$

It is also not too difficult to see that the polynomial in k prefactor in the upper bound on $\text{Tr}(\Pi_\epsilon \Omega)$ above is the largest such prefactor and thus Eq. (130) indeed holds if we take $p(k)$ to be

$$p(k) = 12^3(k\tau_{\max} + 4)^{\frac{n(n-1)}{2}+1}. \quad \text{\textcolor{blue}{\{eq:p_of_k\}}} \quad (144)$$

□

While Theorem 42 states that $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ concentrates in magnitude around particular Young diagrams, it is not directly applicable to the tetrahedral Horn problem simply because its value depends on more than just the eigenvalues of the matrices (A, B, C, D, E, F) . Nevertheless, combining Theorem 42 with the completeness relation (Eq. (124)) yields a necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral. \text{\textcolor{blue}{cor:tet_size_lower_bound}}

Corollary 43. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) with minimum trace $\tau_{\min}$. Then there exists a sextuple of sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by positive integers $k > \tau_{\min}^{-1}$ with sizes given by Eq. (129) satisfying both (i) the convergence condition*

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log p(k)q(k) \xrightarrow{k \rightarrow \infty} 0, \quad \text{\textcolor{blue}{\{eq:sextuple_convergence_rate\}}} \quad (145)$$

and (ii) the lower bound

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\infty} \geq \frac{1}{q(k)}, \quad \text{\textcolor{blue}{\{eq:ud_size_lower_bound\}}} \quad (146)$$

where $p(k)$ is given by Eq. (144) whereas $q(k)$ is given by

$$q(k) = (k\tau_{\max} + 4)^{6(r_{\max}-1)}. \quad \text{\textcolor{blue}{\{eq:q_of_k\}}} \quad (147)$$

Proof. To proof the claim, we construct the sequence implicitly. For each k , it suffices to take $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ to be any sextuple of Young diagrams which maximizes the value of $\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right|$ with sizes determined by Eq. (129):

$$(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k) := \underset{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}{\operatorname{argmax}} \left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right|. \quad (148)$$

While this maximal choice may not be unique, it will necessarily satisfy a lower bound due to the completeness relation satisfied by $\Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix}$ Eq. (124). Specifically, this maximal choice must satisfy

$$\left| \Gamma \begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right| \geq \frac{1}{q(k)} \quad \text{\textcolor{blue}{\{eq:tet_gamma_lower_bound\}}} \quad (149)$$

where $q(k)$ is any upper bound on the number of possible summands in the completeness relation Eq. (124) viewed as a function of k . As there less than or equal to $(s+1)^{r-1}$ Young diagrams with r rows with size at most s , letting $q(k)$ be given by Eq. (147) ensures Eq. (149) holds because the size (resp. rank) of any of $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ is at most $n\tau_{\max} + 3$ (resp. $r_{\max}$). Moreover, applying Eq. (130) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k} \leq \frac{6}{k} \log(q(k)p(k)) \quad (150)$$

and thus Eq. (145) holds because $p(k)$ and $q(k)$ are polynomials in k with coefficients depending on the maximum trace $\tau_{\max}$ and maximum rank $r_{\max}$. Also Eq. (146) follows from Eq. (149) combined with the upper bound provided by Eq. (126). □

Remark 44. The convergence condition above (Eq. (145)) in conjunction with the size conditions given by Eq. (129) together ensure that the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the following sense that

$$\begin{aligned} \frac{\alpha'_k}{k} &\rightarrow \alpha, & \frac{\beta'_k}{k} &\rightarrow \beta, & \frac{\gamma'_k}{k} &\rightarrow \gamma, \\ \frac{\delta'_k}{k} &\rightarrow \delta, & \frac{\epsilon'_k}{k} &\rightarrow \epsilon, & \frac{\phi'_k}{k} &\rightarrow \phi, \end{aligned} \quad \text{\textcolor{blue}{\{eq:sextuple_convergence\}}} \quad (151)$$

where $\frac{\lambda'_k}{k} \rightarrow \lambda$ is shorthand for $\lim_{k \rightarrow \infty} (\lambda'_k)_i / k = \lambda_i$ for all $i \in \{1, \dots, n\}$.

While the existence of a convergent sequence $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ of primed Young diagrams is necessary for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral by Corollary 43, our next result proves it is also sufficient. In other words, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, then no such sequence can exist.

Corollary 45. *Suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is a non-tetrahedral sextuple of positive semidefinite and non-zero eigenvalues with minimum trace $\tau_{\min}$. Then for all sequences of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ indexed by $k > \tau_{\min}^{-1}$ with sizes given by Eq. (129) and converging to $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that Eq. (145) holds, we have*

$$\left\| \left\{ \begin{smallmatrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-\frac{k\tau_{\min}}{12} D_*^{\alpha\beta\gamma\delta\epsilon\phi} + O(\log k) \right), \quad \text{\textcolor{blue}{eq:seq_ud_sizj_upper_bound}} \quad (152)$$

where $D_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ is a positive constant which measures the distance from $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the closed set of tetrahedral eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ with matching traces (i.e., $|\alpha| = |\tilde{\alpha}|$, $|\beta| = |\tilde{\beta}|$, etc.):

$$D_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral} \\ \forall \lambda: |\tilde{\lambda}| = |\lambda|}} \left(\|\underline{\alpha} - \underline{\tilde{\alpha}}\|_1^2 + \|\underline{\beta} - \underline{\tilde{\beta}}\|_1^2 + \|\underline{\gamma} - \underline{\tilde{\gamma}}\|_1^2 + \|\underline{\delta} - \underline{\tilde{\delta}}\|_1^2 + \|\underline{\epsilon} - \underline{\tilde{\epsilon}}\|_1^2 + \|\underline{\phi} - \underline{\tilde{\phi}}\|_1^2 \right). \quad (153)$$

Proof. Using Eq. (118), we have

$$R_{\alpha'\beta'\delta';\epsilon'} \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \iiint d\mu(A) d\mu(B) d\mu(D) \Gamma \left[\begin{smallmatrix} A & B & C \\ D & E & F \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \quad (154)$$

where the integral is taken over a triple of $n \times n$ trace one positive semidefinite matrices A, B, D . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| A, & \tilde{B} &:= |\beta| B, & \tilde{C} &:= |\alpha| A + |\beta| B, \\ \tilde{D} &:= |\delta| D, & \tilde{E} &:= |\alpha| A + |\beta| B + |\delta| D, & \tilde{F} &:= |\beta| B + |\delta| D. \end{aligned} \quad \text{\textcolor{blue}{eq:tilde_defs}} \quad (155)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is also equal to $\Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right]$. Moreover, by Eq. (130) we have a bound on the integrand,

$$\left| \Gamma \left[\begin{smallmatrix} \tilde{A} & \tilde{B} & \tilde{C} \\ \tilde{D} & \tilde{E} & \tilde{F} \end{smallmatrix} \middle| \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right] \right| \leq p(k) \exp(-k I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} / 6). \quad (156)$$

Now by construction, the eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral while by assumption, the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral (although they have matching traces due to Eq. (155)). Therefore we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (157)$$

and therefore $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi} > 0$.

Our goal now will be to show that for any sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converging to the non-tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, one can also guarantee that for sufficiently large k , the exponent $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ above is also bounded from below by a positive constant. Let λ be a placeholder variable indexing $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$. By Pinsker's inequality for the relative entropy, we obtain a sequence of lower bounds on $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$:

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \sum_{\lambda} |\lambda| D(\underline{\lambda}' \| \underline{\lambda}) \geq \frac{1}{2} \sum_{\lambda} |\lambda| \|\underline{\lambda}' - \underline{\lambda}\|_1^2 \geq \frac{\tau_{\min}}{2} \sum_{\lambda} \|\underline{\lambda}' - \underline{\lambda}\|_1^2 \geq \frac{\tau_{\min}}{2} \max_{\lambda} \|\underline{\lambda}' - \underline{\lambda}\|_1^2, \quad (158)$$

where $\tau_{\min}$ is the minimum trace over the eigenvalues (the lower indices). Applying the last bound to $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ yields

$$\max_{\lambda} \|\underline{\lambda}'_k - \underline{\lambda}\|_1 \leq \sqrt{2\tau_{\min}^{-1} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} \quad (159)$$

whereas applying the second to last bound to $I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ (noting that $\tau_{\min}$ is also the minimum trace of $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$) yields

$$\sum_{\lambda} \left\| \underline{\lambda}'_k - \tilde{\lambda} \right\|_1^2 \leq 2\tau_{\min}^{-1} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}. \quad (160)$$

Furthermore, by the reverse triangle inequality we have

$$\sum_{\lambda} \left\| \underline{\lambda}'_k - \tilde{\lambda} \right\|_1^2 \geq \sum_{\lambda} \left(\left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 - \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1 \right)^2 \geq \sum_{\lambda} \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1^2 - 2 \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1. \quad (161)$$

Also by the definition of $D_*^{\alpha\beta\gamma\delta\epsilon\phi}$ we have

$$D_*^{\alpha\beta\gamma\delta\epsilon\phi} \leq \sum_{\lambda} \left\| \underline{\lambda} - \tilde{\lambda} \right\|_1^2. \quad (162)$$

Since $\underline{\lambda}$ and $\tilde{\lambda}$ are both probability distributions we also have $\left\| \underline{\lambda} - \tilde{\lambda} \right\|_1 \leq 2$. Also, $6 \max_{\lambda} \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1 \geq \sum_{\lambda} \left\| \underline{\lambda}'_k - \underline{\lambda} \right\|_1$ as there are six summands. Combining all these bounds together yields

$$\left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \exp \left(-k \left\{ \frac{\tau_{\min} D_*^{\alpha\beta\gamma\delta\epsilon\phi}}{12} - 2\sqrt{2\tau_{\min} I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}} - \frac{1}{k} \log(R_{\alpha'_k, \beta'_k, \delta'_k, \epsilon'_k}^{-1} p(k)) \right\} \right). \quad (163)$$

By assumption, the Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ converge in normalization to the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ in the sense that $I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_k\beta'_k\gamma'_k\delta'_k\epsilon'_k\phi'_k}$ tends to zero as $k \rightarrow \infty$ at a rate given by Eq. (145). Moreover, $R_{\alpha'_k, \beta'_k, \delta'_k, \epsilon'_k}^{-1}$ can be upper bounded by a polynomial in k and $p(k)$ is a polynomial in k so Eq. (152) holds as claimed. $\square$

4.4 The large n limit & an entropic inequality

sec:large_n

The purpose of this section is to show how to use the necessary condition for $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to be tetrahedral provided by Corollary 43 to derive inequalities satisfied by tetrahedral eigenvalues. Our strategy will be to prove an upper bound the generalized $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 43, we obtain a new constraint on the convergent sequence of Young diagrams $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$.

thm:sixj_ssa_bound

Theorem 46. *Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be a sextuple of Young diagrams. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 \leq \left(c_{\alpha', \phi'}^{\epsilon'} c_{\beta', \delta'}^{\phi'} c_{\alpha', \beta'}^{\gamma'} c_{\gamma', \delta'}^{\epsilon'} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)}. \quad (164)$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, V_{λ}^n , and omit the multiplicity space labels on the intermediate steps.

$$\left(\begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} \right) = \left(\begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} \right) \leq \left[\begin{array}{c} \text{Diagram 5} \\ \text{Diagram 6} \end{array} \right]^{\frac{1}{2}} = \frac{\left(\begin{array}{c} \epsilon \\ \gamma \\ \phi \\ \beta \end{array} \right)}{\left(\begin{array}{c} \beta \\ \gamma \\ \epsilon \end{array} \right)} \left[\begin{array}{c} \text{Diagram 7} \\ \text{Diagram 8} \\ \text{Diagram 9} \\ \text{Diagram 10} \end{array} \right]^{\frac{1}{2}} \quad (165)$$

The first equality follows from Definition 38. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying

Schur's lemma to components of the diagram mapping $V_{\beta'}^n$ to itself on the left and $V_{\epsilon'}^n$ to itself on the right. Note that the factors of $\dim(V_{\epsilon'}^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Theorem 46 is to combine it with Corollary 43. If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues, then by Corollary 43 there exists a converging sequence of Young diagrams $(\alpha'_k, \beta'_k, \gamma'_k, \delta'_k, \epsilon'_k, \phi'_k)$ such that⁸

$$\frac{1}{q(k)^2} \leq \left\| \left\{ \begin{matrix} \alpha'_k & \beta'_k & \gamma'_k \\ \delta'_k & \epsilon'_k & \phi'_k \end{matrix} \right\} \right\|_{\text{HS}}^2 \leq \left(c_{\alpha'_k, \phi'_k}^{\epsilon'_k} c_{\beta'_k, \delta'_k}^{\phi'_k} c_{\alpha'_k, \beta'_k}^{\gamma'_k} c_{\gamma'_k, \delta'_k}^{\epsilon'_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma'_k}^n) \dim(V_{\phi'_k}^n)}{\dim(V_{\beta'_k}^n) \dim(V_{\epsilon'_k}^n)}. \quad (166)$$

Remarkably, both $q(k)$ (given by Eq. (147)) and the Littlewood-Richardson coefficients $c_{\alpha'_k, \phi'_k}^{\epsilon'_k}, c_{\beta'_k, \delta'_k}^{\phi'_k}, c_{\alpha'_k, \beta'_k}^{\gamma'_k}$, and $c_{\gamma'_k, \delta'_k}^{\epsilon'_k}$ are *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams, i.e., $n \geq r_{\max}$ where $r_{\max}$ is the maximum rank of any of the tetrahedral eigenvalues). This observation provides one with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_{\lambda'}^n)$, of the irreducible representation $V_{\lambda'}^n$ of $U(n)$ grows as n becomes much larger than the number of rows of λ' . In general, for fixed λ' , the function $n \mapsto V_{\lambda'}^n$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_{\lambda'}^n)$ in the regime where n is large compared to $|\lambda'|$ and also the last row $\lambda'_{\min}$ is large compared to the depth of λ' in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, namely [Har23, Lemma 2], which relates the asymptotics of $\dim(V_{\lambda'}^n)$ to $\dim(W_{\lambda'})$. prop:harrow_formula

Proposition 47. *Let λ' be a Young diagram with depth at most n . Then*

$$\frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!} \prod_{(i,j) \in \lambda'} \left(1 + \frac{j-i}{n} \right). \quad (167)$$

Proof. The statement follows from the dimension formulas for $\dim(V_{\lambda'}^n)$ and $\dim(W_{\lambda'})$ in Eq. (91) and Eq. (92). See also [Har23, Appendix A]. $\square$

From Proposition 47 we obtain the following corollary. cor:harrow_ssa_lim

Corollary 48. *Let $\gamma', \phi', \beta', \epsilon'$ be a quadruple of Young diagrams such that*

$$|\gamma'| + |\phi'| = |\beta'| + |\epsilon'|. \quad (168)$$

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\gamma'}^n) \dim(V_{\phi'}^n)}{\dim(V_{\beta'}^n) \dim(V_{\epsilon'}^n)} = \frac{|\beta'|! |\epsilon'|! \dim(W_{\gamma'}) \dim(W_{\phi'})}{|\gamma'|! |\phi'|! \dim(W_{\beta'}) \dim(W_{\epsilon'})} \quad (169)$$

Proof. From Proposition 47 one evidently has for any Young diagram λ' the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\lambda'}^n)}{n^{|\lambda'|}} = \frac{\dim(W_{\lambda'})}{|\lambda'|!}. \quad (170)$$

The claim then follows because the missing prefactor, $\frac{n^{|\beta'|} n^{|\epsilon'|}}{n^{|\gamma'|} n^{|\phi'|}}$, is equal to one by Eq. (168). $\square$

We are now in a position to prove the main result of this section.

Theorem 49. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be tetrahedral eigenvalues. Then*

$$I(\gamma, \phi; \beta, \epsilon) = s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \geq 0, \quad (171)$$

where

$$s(\lambda) := - \sum_{i=1}^n \lambda_i \log \lambda_i. \quad (172)$$

⁸Here we used the well-known inequality $\|X\|_{\infty} \leq \|X\|_{\text{HS}}$.

Proof. As $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral, the trace conditions (Eq. (12)) necessarily imply that Eq. (168) holds. Therefore, by Corollary 48 and the discussion surrounding the bound in Eq. (166) we conclude that along the convergent sequence of Young diagrams satisfies

$$I' := \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[\frac{|\beta'_k|! |\epsilon'_k|! \dim(W_{\gamma'_k}) \dim(W_{\phi'_k})}{|\gamma'_k|! |\phi'_k|! \dim(W_{\beta'_k}) \dim(W_{\epsilon'_k})} \right] \geq 0 \quad (173)$$

provided the limit I' exists. An application of Stirling's approximation [Rob55] to the dimension formula for $\dim(W_{\lambda'_k})$ together with the approximation $\lambda'_k \approx k\lambda$ yields

$$\frac{1}{k} \frac{\dim(W_{\lambda'_k})}{|\lambda'|!} = s(\lambda) + |\lambda| (1 - \log k) + O(k^{-1} \log k) \quad (174)$$

where $s(\lambda)$ is given in Eq. (172). Moreover by Eq. (168), the $(1 - \log k)$ terms cancel and therefore $I' = I(\gamma, \phi; \beta, \epsilon) \geq 0$ as claimed above. $\square$

A Multinomial distribution inequalities

lem:multinomial_dist_lower_bound

Lemma 50. Fix $m \geq 2$ and let $(w_1, \dots, w_m) \in \mathbb{R}_{>0}^m$ have minimum $w_{\min} := \min\{w_1, \dots, w_m\} > 0$. Let k be a positive integer satisfying $k \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_m) \in \mathbb{N}_{>0}^m$ by the equation

$$z_i := \lceil kw_i \rceil. \quad (175)$$

Let $Z = z_1 + \dots + z_m$ and $W = w_1 + \dots + w_m$. Then

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{12} \right)^m (kW + m)^{\frac{m-1}{2}} \quad \text{\textit{feg:multinomial_dist_lower_bound}} \quad (176)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$k! = \sqrt{2\pi} k^{k+\frac{1}{2}} e^{-k} e^{r_k}, \quad \text{where} \quad \frac{1}{12k+1} \leq r_k \leq \frac{1}{12k}. \quad (177)$$

Therefore,

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} = \left(\frac{1}{2\pi Z} \right)^{\frac{m-1}{2}} \prod_{i=1}^m \left(\frac{Z}{z_i} \right)^{\frac{1}{2}} \left(\frac{Z w_i}{z_i W} \right)^{z_i} \exp \left(r_Z - \sum_{i=1}^m r_{z_i} \right). \quad (178)$$

In this setting, we have let $z_i = \lceil kw_i \rceil$ which implies

$$kw_i \leq z_i < kw_i + 1, \quad \text{and} \quad kW \leq Z < kW + m. \quad (179)$$

Consequently⁹,

$$\left(\frac{Z w_i}{z_i W} \right)^{z_i} \geq \left(\frac{kw_i}{kw_i + 1} \right)^{kw_i + 1} \geq \frac{1}{4}. \quad (180)$$

The remainder term satisfies

$$r_Z - \sum_{i=1}^m r_{z_i} \geq \frac{1}{12Z+1} - \sum_{i=1}^m \frac{1}{12z_i} \geq -\frac{m}{12}. \quad (181)$$

Also $Z \geq z_i$ so altogether, we have

$$\frac{Z!}{z_1! \dots z_m!} \frac{w_1^{z_1} \dots w_m^{z_m}}{W^Z} \geq \left(\frac{1}{2\pi} \right)^{\frac{m-1}{2}} \left(\frac{1}{4} \right)^m \exp \left(-\frac{m}{12} \right) \left(\frac{1}{kW+m} \right)^{\frac{m-1}{2}} \geq \left(\frac{1}{12} \right)^m (kW+m)^{\frac{m-1}{2}}, \quad (182)$$

as claimed because $\sqrt{2\pi} > 1$ and $\frac{1}{12} < \frac{e^{-\frac{1}{12}}}{4\sqrt{2\pi}} = 0.09176 \dots$. $\square$

⁹Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1} \right)^{x+1} \geq \frac{1}{4}$.

B A character formula for $6j$ symbols

Throughout Section 4.3 we proved a number of results concerning the square of the Hilbert-Schmidt norm of the $6j$ operator defined by Definition 38. The following result provides a formula for the square of the Hilbert-Schmidt norm as an integral over four copies of $U(n)$.

lem:character_formula

Lemma 51 (Character formula). *Let $\left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} : C_{\alpha',\beta'}^{\gamma'} \otimes C_{\delta',\gamma'}^{\epsilon'} \rightarrow C_{\alpha',\phi'}^{\epsilon'} \otimes C_{\beta',\delta'}^{\phi'}$ be the $6j$ operator defined in Definition 38. Then*

$$\dim(V_{\epsilon'})^2 \left\| \left\{ \begin{smallmatrix} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \int_{U(n)^4} du_1 du_2 du_3 du_4 X_{\delta',\epsilon',\phi'}^{\alpha',\beta',\gamma'}(u_1, u_2, u_3, u_4). \quad (183)$$

eq:character_formula

where

$$X_{\delta',\epsilon',\phi'}^{\alpha',\beta',\gamma'}(u_1, u_2, u_3, u_4) := \chi_{\alpha'}(u_1^{-1}u_2) \chi_{\beta'}(u_4^{-1}u_2) \chi_{\gamma'}(u_2^{-1}u_3) \chi_{\delta'}(u_4^{-1}u_3) \chi_{\epsilon'}(u_3^{-1}u_1) \chi_{\phi'}(u_1^{-1}u_4). \quad (184)$$

Note that when $n = 2$, Eq. (183) recovers the well-known formula for the square of the $6j$ -symbol for $SU(2)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of Lemma 51. The proof follows from (i) the following identities concerning the character $\chi_{\lambda'}(u) = \text{Tr}(\pi_{\lambda'}(u))$ and duality $\pi_{\lambda'}^*(u) = (\pi_{\lambda'}(u^{-1}))^T$ relations

$$\chi_{\lambda}(u) = \boxed{\pi_{\lambda}(u)} \quad \pi_{\lambda}^*(u) = \boxed{\pi_{\lambda}(u^{-1})}, \quad (185)$$

and (ii) the integral formula for the projection operator $\Pi_{\alpha,\beta}^{\gamma}$ onto the subspace of invariant vectors in $V_{\alpha} \otimes V_{\beta} \otimes V_{\gamma}^*$:

$$\Pi_{\alpha,\beta}^{\gamma} = \boxed{\Pi_{\alpha,\beta}^{\gamma}} = \int du \boxed{\pi_{\alpha}(u)} \boxed{\pi_{\beta}(u)} \boxed{\pi_{\gamma}^*(u)}. \quad (186)$$

Applied four times, we obtain Eq. (183):

$$\begin{aligned} & \left(\begin{array}{c} \boxed{\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix}} \\ C_{\alpha\phi}^{\epsilon} \quad C_{\beta\delta}^{\phi} \\ \boxed{\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix}} \\ C_{\alpha\beta}^{\gamma} \quad C_{\gamma\delta}^{\epsilon} \end{array} \right)^2 = \begin{array}{c} \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \epsilon \quad \delta \quad \delta \quad \beta \quad \phi \\ C_{\alpha\phi}^{\epsilon} \quad C_{\alpha\beta}^{\gamma} \quad C_{\gamma\delta}^{\epsilon} \quad C_{\beta\delta}^{\phi} \\ \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \epsilon \quad \delta \quad \delta \quad \beta \quad \phi \\ C_{\alpha\phi}^{\epsilon} \quad C_{\alpha\beta}^{\gamma} \quad C_{\gamma\delta}^{\epsilon} \quad C_{\beta\delta}^{\phi} \end{array} \\ & = \int du_1 du_2 du_3 du_4 \begin{array}{c} \bar{u}_1 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_4 \quad u_2 \quad \bar{u}_4 \quad u_1 \quad \bar{u}_3 \quad \bar{u}_1 \quad u_4 \\ \alpha \quad \gamma \quad \delta \quad \beta \quad \epsilon \quad \phi \end{array} \end{aligned} \quad (187)$$

□

We conclude by noting that the integrands appearing in the above character formula are closely related to the character formula for the heat kernel of a compact Lie group G [Feg78], denoted by $K_G(u_1, u_2; t)$ in [McS21]:

$$K_G(u_1, u_2; t) = \sum_{\lambda'} \dim V_{\lambda'}^n \chi_{\lambda'}(u_1 u_2^{-1}) e^{-t(|\lambda' + \rho|^2 - |\rho|^2)/2}, \quad (188)$$

where the sum is over all dominant integral weights λ' for G and ρ the associated Weyl vector. It is conceivable that the square of the Hilbert-Schmidt norm of the generalized $6j$ symbols for G can be seen as a low temperature limit, $t \rightarrow 0$, of an integral over products of heat kernels.

C Seesaw reciprocity & eigenvalues of blocks

In Section 4.2, we reviewed the correspondence between (i) the existence of $n \times n$ positive semidefinite matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$ and (ii) the positivity of corresponding Littlewood-Richardson coefficients $c_{\alpha\beta}^\gamma$ which govern the branching problem for the diagonal subgroup $\mathrm{GL}(n) \hookrightarrow \mathrm{GL}(n) \times \mathrm{GL}(n)$ sending X to (X, X) . Recalling Theorem 17, there is a further correspondence to (iii) the existence of a $2n \times 2n$ positive semidefinite matrix Σ with eigenvalues γ and $n \times n$ blocks A', B' with eigenvalues α, β . One may wonder how to show the correspondence between (ii) and (iii) *directly* without passing through (i). This can be accomplished by considering the branching multiplicities $m_\gamma^{\alpha\beta}$ for the blockwise subgroup $\mathrm{GL}(a) \times \mathrm{GL}(b) \hookrightarrow \mathrm{GL}(a+b)$ sending (A, B) to $A \oplus B$. In fact there exists a representation-theoretic analogue of Theorem 17, specifically [GW00, Theorem 9.2.3]. In particular, the values of $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ coincide provided the Young diagrams (α, β, γ) involved have depths at most $(\min\{a, n\}, \min\{b, n\}, \min\{a+b, n\})$; this ensures that $V_\alpha^a, V_\beta^b, V_\gamma^{a+b}$ and $V_\alpha^n, V_\beta^n, V_\gamma^n$ are all non-zero representation spaces and therefore $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ are both meaningfully defined. The distinction between $m_\gamma^{\alpha\beta}$ and $c_{\alpha\beta}^\gamma$ lies with the fact that it is possible for $c_{\alpha\beta}^\gamma$ to be zero while $m_\gamma^{\alpha\beta}$ is non-zero (this can happen when the depth of γ is greater than n but at most $a+b$ so that V_γ^n is zero but V_γ^{a+b} is non-zero). This parallels the fact that, in general, the rank of the $2n \times 2n$ matrix Σ in Theorem 17 could be greater than n while the rank of the $n \times n$ matrix C is at most n .

TODO: Describe the same correspondence for the blockwise $6j$ symbols.

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